INSTRUCTIONS

GEK-36519

STATIC VOLTAGE REGULATOR

FOR

AC GENERATOR

These instructions *do* not purport to cover *all* details or *variations* in equipment nor *to* provide for every possible contingency *to* be met in connection with *installation,* operation *or* maintenance. *Should* further information be desired or should particular problems arise which ore *not* covered sufficiently for the purchaser's *purposes, fhe matter should be* referred *to General* Electric *Company.*

GENERAL@ ELECTRIC

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STATIC VOLTAGE REGULATOR

FOR

AC GENERATOR.

INTRODUCTION

The 3S7930SA112A1, 113A1 voltage regulator controls the voltage of a 60 cycle AC venerator by controlling the field current to the generator field. 'A silicon controlled rectifier controls power delivered to the generator field. The regulator includes a current transformer forcing unit to provide excitation: thus, sustains short circuit current for selective relay tripping, where the voltage regulator input fails due to a fault on the system. The regulator is completely static, having no moving parts to perform the regulating function.

RECEIVING, HANDLING AND STORAGE

Receiving and Handling

Immediately upon receipt, the equipment should be carefully unpacked to avoid damage:, Particular care should be exercised to prevent small parts from being mislaid or thrown away in the packing material.

As soon as the equipment is unpacked, it should be examined for any damage that may have been sustained in transit. If injury or rough handling is evident, a damage claim should be filed immediately with the transportation company and the nearest General Electric Sales should be notified promptly.

Storage

If the equipment is not to be used after unpacking, it should be stored in a clean, dry, place and protected from accidental damage. Particular care should be exercised to avoid storing the equipment in locations where construction work is in progress.

DESCRIPTION

The 3S7930SA112A1, 113A1 voltage regulator consists of a basic regulator, a current transformer forcing unit, a droop kit and miscellaneous components.

The basic voltage regulator consists of a regulator panel assembly (including terminal board and adjustments) with rectifiers and heat sinks mounted on the supporting steel base.

The voltage adjusting rheostat is furnished mounted on the base.

The regulator includes a current transformer forcing unit. This unit is to provide excitation and sustained short circuit current for selective relay tripping during generator faults.

The regulator includes a droop kit for dividing reactive current when paralleling generators.

The regulator also includes necessary transformers to provide operation from a 400 to 480 volt generator.

INSTALLATION

The regulator should be mounted so that it is accessible. It should be mounted in an enclosure to protect personnel from exposed voltages. The enclosure should allow a reasonable circulation of air to keep the ambient temperature at 50°C or lower. The regulator must be mounted in an upright position as indicated on the outline drawing. The current transformers for C.T. forcing, are to be remotely mounted by user. The current transformer for the droop kit (furnished by others) is also remotely mounted.

Make all connections to the regulator per diagram furnished with equipment. Connect all remotely mounted components per diagram furnished with equipment.

OPERATION

WARNING: THE HEAT SINKS ON THIS REGULATOR ARE AT ABOVE GROUND POTENTIAL.

CAUTION: TO PREVENT DAMAGE TO THE EXCITER OR GENERATOR, THE PRIME MOVER SHOULD NOT BE OPERATED AT LESS THAN 75% SPEED FOR MORE THAN 30 SECONDS WITH THE REGULATOR ON.

CAUTION: TO PREVENT DAMAGE TO REGULATOR, EXCITER FIELD RESISTANCE SHOULD NOT BE LESS THAN 0.8 OHMS.

CAUTION: ALWAYS OPEN CURRENT FLOW THROUGH 3CT PRIMARY BEFORE CHANGING TAPS OR SHORT 3CT SECONDARY BEFORE CHANGING TAPS.

Adjustments

Before starting prime mover, set "Voltage Adjust" (1P) and "Stabilizing" (3P) at their mid-points. Set "Gain (2P) fully counterclockwise.

Start prime mover and bring up to rated speed. Set "Voltage Adjust" to obtain proper line voltage..

Turn "Gain" clockwise to improve regulation as generator is loaded. "Gain" should be set to allow approximately 1% voltage drop going from a no-load to a full-load condition on the generator. It will be necessary to correct voltage level with the "Voltage Adjust" each time the "Gain" is changed.

To obtain optimum transient response when applying and rejecting load, vary the "Stabilizing" (3P) and the connections on 7TB. The "Stabilizing" adjust (3P) acts as a fine adjustment; while the connections on 7TB act as a course or range adjustment.

Start prime mover and bring up to speed. Flash generator field if this is required and flashing is not already initiated by automatic relaying. Set 1P to obtain proper generator voltage.

To obtain optimum response when applying and rejecting load, vary "Stability" (3P) and the capacity at 7TB. Starting with maximum capacity, adjust 3P for best response as the capacity is decreased in steps.

Too much capacity will cause the system to be over-damped, while not enough capacity will cause the system to be under-damped. See 'Figure 1. Adjusting 3P will give optimum response for each step of changing capacity. System may be unstable at either end of 3P adjustment, In effect, varying capacity with jumpers at 7TB is a coarse stabilizing adjustment while varying 3P is a fine stabilizing adjustment.

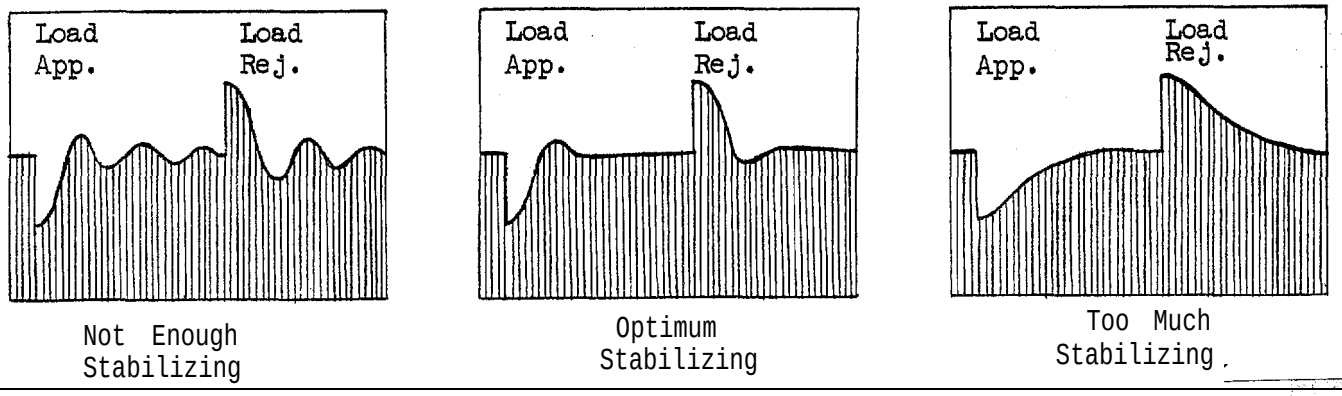
With stabilizing circuit near optimum, increase the "Gain" (2P) until the generator voltage drops between 0.1% and 0.5% as the generator changes from a no-load to a rated load condition. "Droop" compensation circuit must be "Off" for this steady state regulation check, "Voltage Adjust" (1P)" will need to be reset as "Gain" (2P) is increased.

Go back and recheck "Stability" (3P) for optimum adjustment after "Gain" has been set.

Figure 1 shows the effect of "Stabilizing" adjustment as seen on an oscilloscope or recorder when applying and rejecting loads.

It should not be necessary to test the current transformer forcing unit. The setting of the current transformer on the current transformer forcing unit should have already been arrived at analytically as described in "Principles of Operation".

"Voltage Adjust" should be checked weekly, or monthly as required.



Typical oscillograms looking at an amplified portion of generator ac voltage

Figure 1

PRINCIPLES OF OPERATION

General

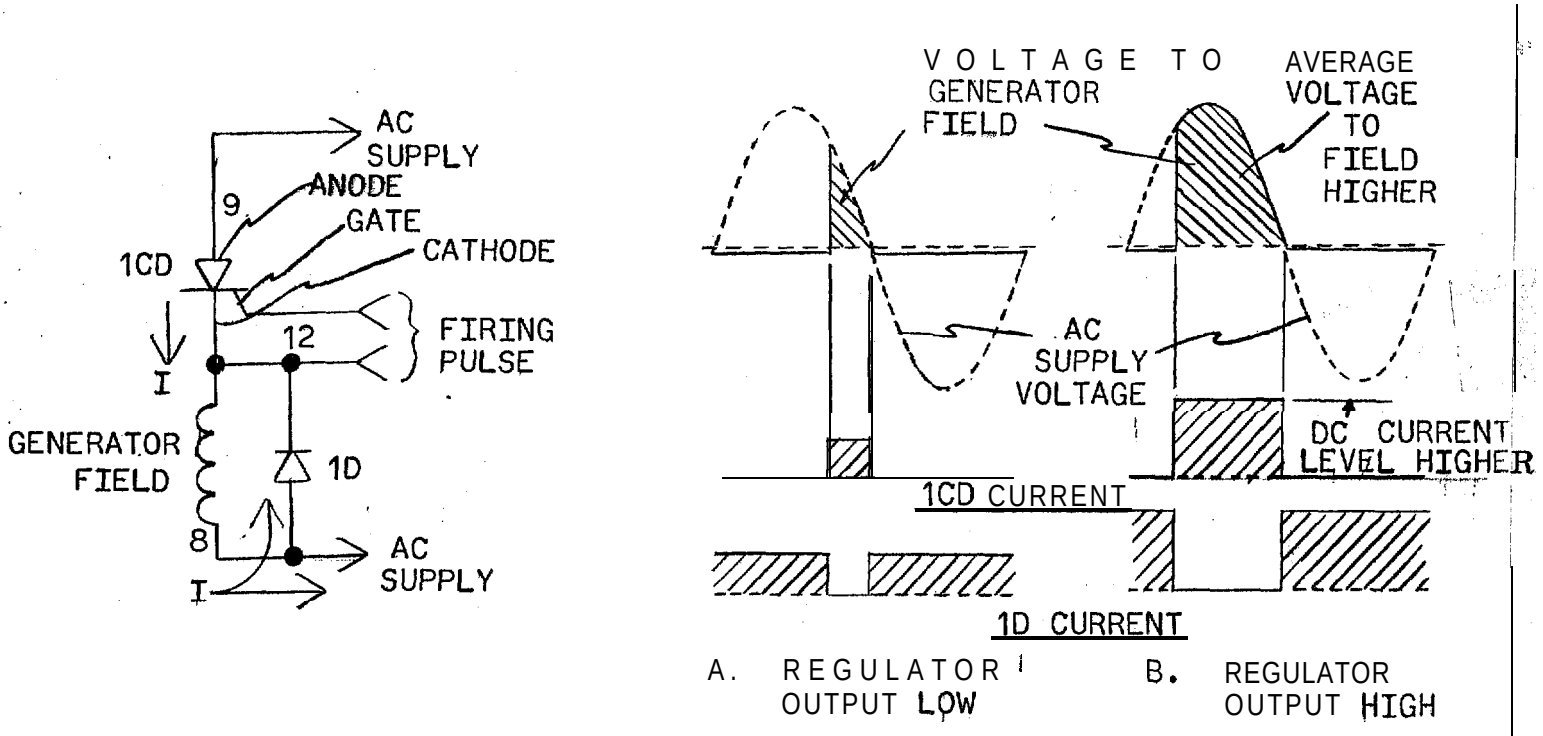
The 3S7930SA112A1, 113A1 voltage regulator operates by taking power from the generator output. and rectifying and controlling the power to furnish dc excitation to the generator field. Refer to Figure 8, the Elementary Diagram for this regulator.

Silicon Controlled Rectifier

The dc excitation power for the exciter field is furnished by single phase half-wave silicon controlled rectifier (SCR). The SCR (1CD) is phase controlled'to control the power delivered to the generator field.

An SCR is similar to a Thyratron tube in that it remains non-conductive until a firing pulse is applied through its gate-to-cathode junction. If the anode is positive with respect to the cathode when a pulse is applied, the SCR will conduct and remain conducting until the anode goes negative and the current goes to zero,

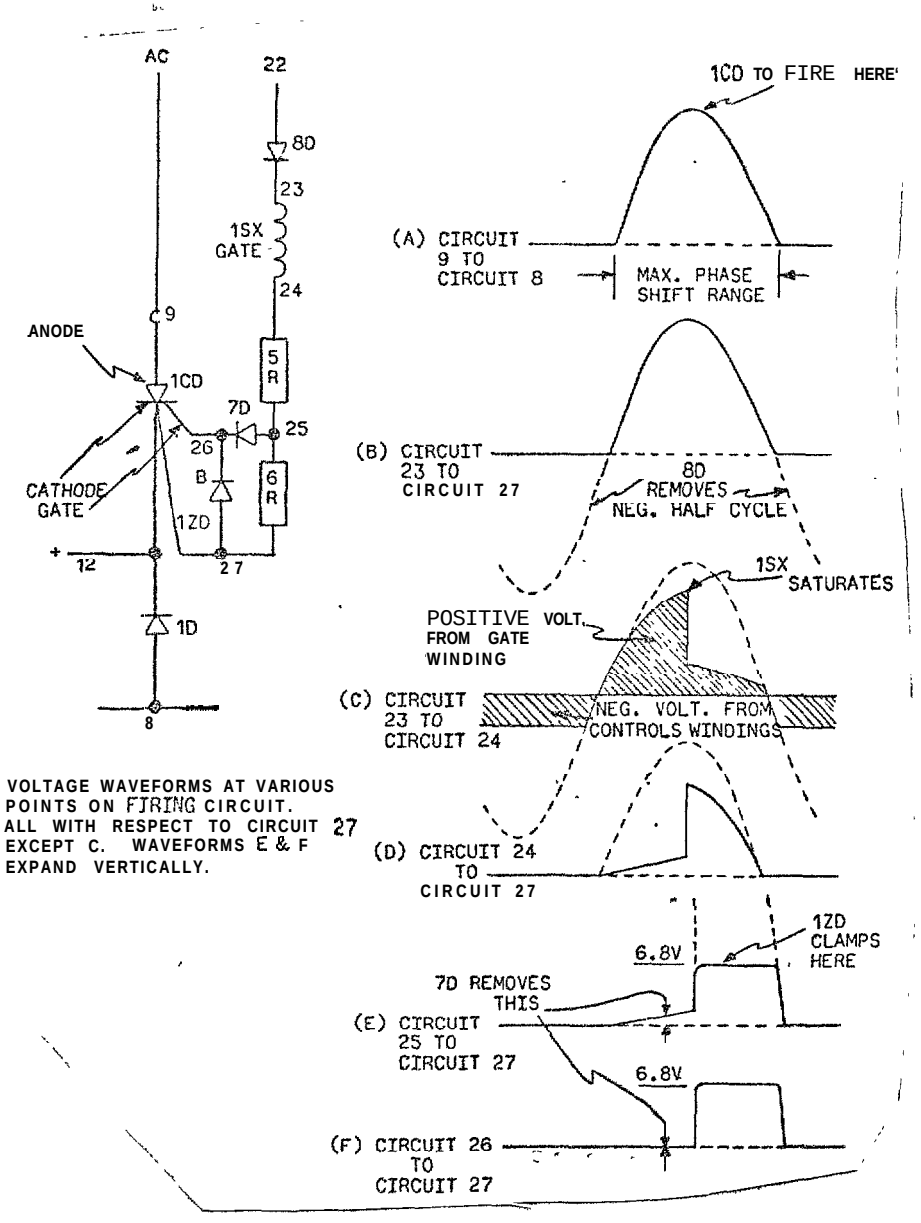
The voltage regulator controls the generator field voltage by -controlling the firing of 1CD on each positive half cycle. See Figure 2 .



Phase Controlled Output of SCR (1CD)

Figure 2

If 1CD fires early in each positive half cycle (Figure 2B), the regulator delivers a larger average voltage to the generator field. If 1CD fires late in each positive half cycle (Figure 2A), the regulator delivers the smaller average voltage to the exciter field. Although the voltage applied to the generator field is portions of a sine wave, the current through the field is continuous because of the inductance of the field winding. Since 1CD cannot supply voltage from the AC supply during the negative half cycle, the field current conducts through the "free-wheeling" rectifier 1D.



SCR Firing Circuit

Figure 3

Firing Circuit

The typical firing circuit (for 1CD) consists of 8D, 1SX, 5R, 6R, 7D, and 1ZD. See Figure 3. Figure 3(a) shows 1CD anode voltage, indicating the range of time where its anode is positive and it could be fired to conduct in the circuit shown in Figure 2. The L3 and L1 voltage falls at the proper phase to cover the required range of phase shift. Rectifier 8D allows only the positive half-cycle of voltage to be applied to the firing circuit.

During the positive half-cycle of voltage, reactor 1SX will accumulate volt-seconds. Since its exciting current is low, very little voltage will appear across 5R and 6R. The small voltage that does appear across 6R (less than 1/2 volt) is blocked by 7D and does not appear on 1CD gate.

When 1SX has accumulated enough volt-seconds to saturate (depending on how much the flux was reset on the previous negative half cycle) its reactance drops sharply and current through 5R and 6R increases sharply. The increased current transfers voltage to 5R and 6R. The voltage across 6R rises to 6.8 volts (limited by 1ZD); thus, voltage is applied to the gate to cathode junction of the SCR. The SCR will fire and conduct anode current until its anode voltage goes negative.

The point on the 1A and 2A positive half-cycle where 1SX saturates and SCR firing occurs, is determined by how much the control winding has reset the flux on the preceding negative half-cycle. If the control winding negative voltage is low, the flux will be reset very little; thus, 1SX will saturate early on the positive half-cycle when voltage is reapplied. If the control winding negative voltage is high, the flux will be reset much more, thus, 1SX will not accumulate enough volt-seconds to saturate until late in the positive half-cycle. When negative voltage applied to any control winding is increased, SCR firing is retarded and the regulator output decreases.

Wave Shape Filter

Reactor 1X, capacitor 1C, 2C, and 3C and resistor 3R is a filter to prevent the exciter from distorting the generator wave shape. Due to the fast switching characteristics of SCR's, the high rate of change of current can introduce a spike or notch on the generator voltage wave. The filter suppresses this spike or notch,

AC Regulator

The AC regulator provides normal automatic operation by regulating the AC generator voltage. The AC regulator controls the regulator power rectifier circuit by controlling the reset voltage applied to 1SX control windings (8 to 6). See typical elementary diagram - Figure 8.

GEK-36519 Static Voltage Regulator for AC Generators

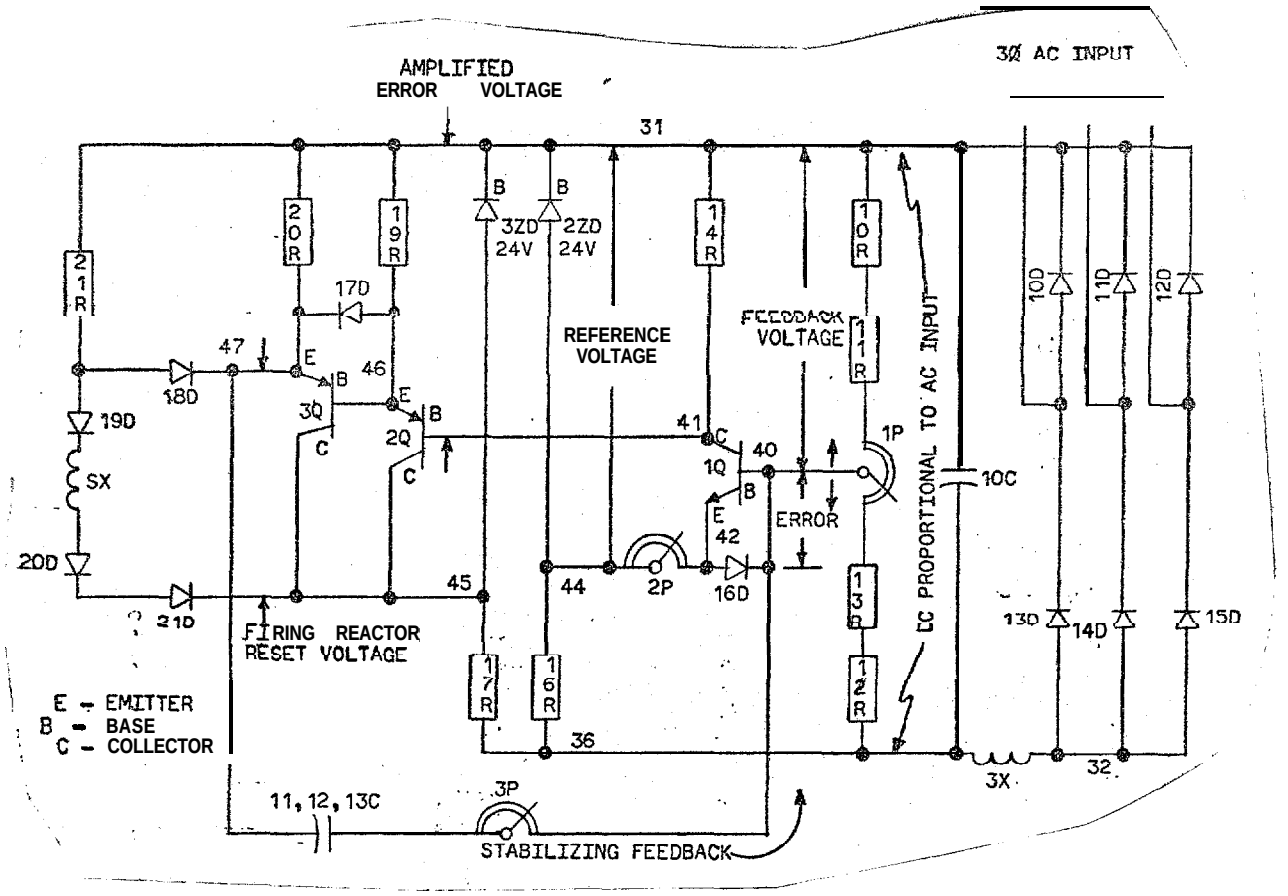
The AC voltage into the regulator must be 400-480 volt level. Transformers 1T and 2T reduce input AC voltage to approximately 42 volts. This voltage is rectified by 10D through 15D and filtered by 3X and to produce a DC voltage across the sensing circuit (31 to 36) that is proportional to the generator AC voltage. See Figure 4. Some models may use. single phase sensing instead of 3 phase sensing as described.

The sensing circuit supplies power through 16R to zener diode 2ZD, providing a 24 volt DC regulated supply across circuits 44 to 31. The sensing circuit also supplies power through 17R to zener diode 3ZD to establish a 24 volt reference voltage for--the regulator. The sensing circuit also supplies'power to the voltage divider consisting of 10, 11, 12, 13R and 1P. This voltage divider provides the feedback voltage across circuits 31 to 40.

The difference between the reference voltage and feedback voltage produces an error current that flows through the base-to-emitter of transistor 1Q. This error current is amplified by 1Q; thus, any small change in 1Q base current produces a large change in 1Q collector current. This amplified error current produces an amplified error signal across 14R, thus setting the voltage level of 2Q base with respect to circuit 31. Transistor 2Q is an emitter follower; therefore, circuit 46 follows the voltage level of circuit 41. Transistor 3Q is also an emitter follower; therefore, circuit 47 also follows circuit 41. By controlling the voltage across circuits 31 to 47, the voltage across circuits 47 to 44 is also controlled, but in the opposite sense. The voltage across 3Q (circuits 47 to 45) sets-the level of voltage available to reset the flux in firing reactor 1SX; thus, controls the phase shift firing of the regulator power rectifier circuit.

Should AC generator voltage dip due to load application: sensing circuit voltage across circuits 31 to 36 will decrease; feedback voltage across circuits 31 to 40 will decrease; error current through 1Q base-to-emitter will increase; 1Q collector current will increase; voltage across 14R will increase: voltage across circuits 41 to 45 will decrease; voltage across circuits 46 and 47 to 45 will decrease; thus, the firing reactors will have less reset voltage available so the power rectifier circuit will turn-on to increase generator excitation and return the AC voltage to normal.

During a change such as just described, the stabilizing circuit (47 to 40) will feed back the changing output voltage to transiently oppose the initial voltage dip and allow the system to return to normal voltage smoothly without oscillating. Should AC generator voltage rise due to load rejection, the opposite of the above action would take place.



AC Regulator

Figure 4

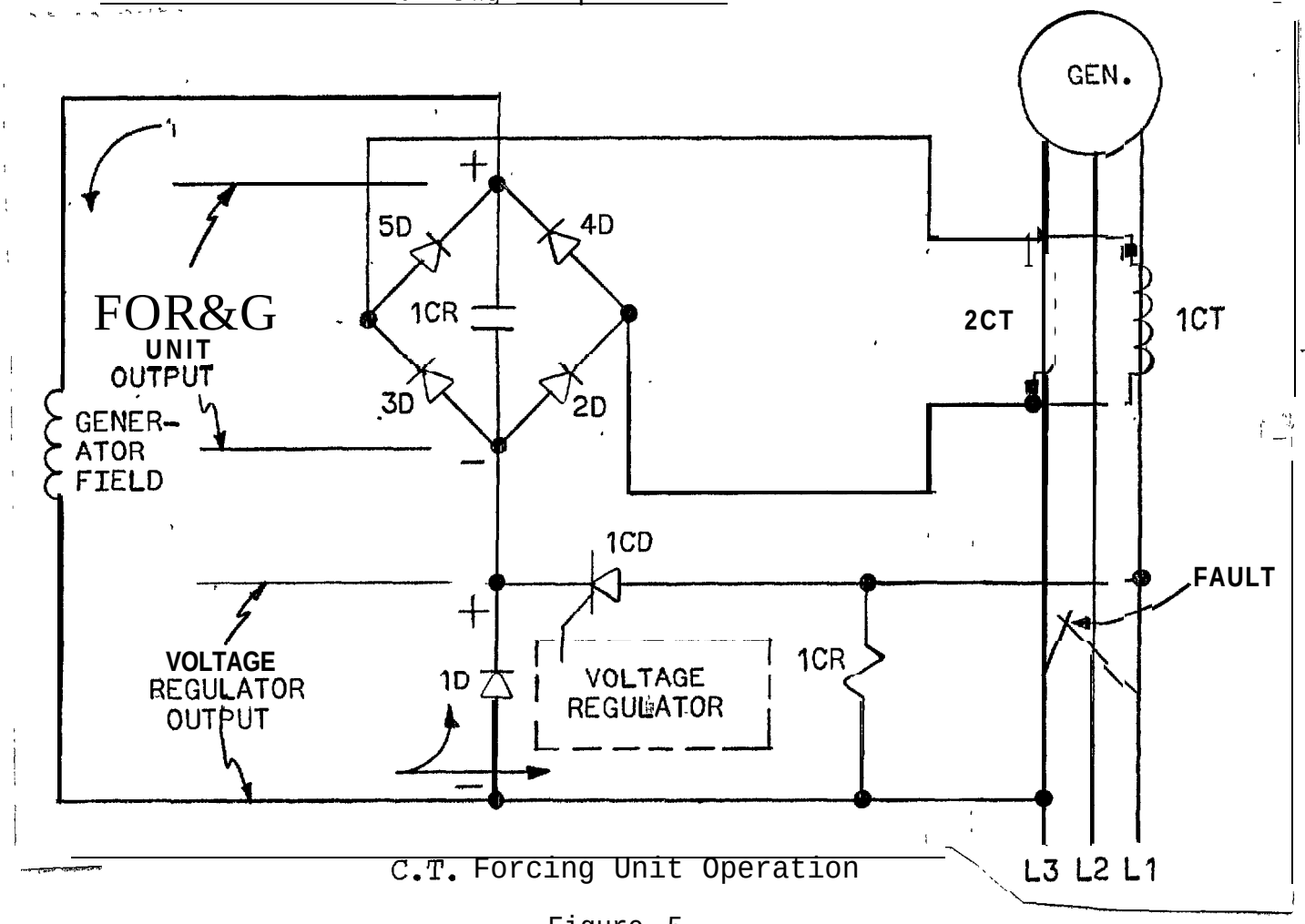
Rectifier 16D shunts any reverse error current around transistor 1Q to protect its emitter-to-base junction. The "Gain" adjust (2P) introduces degeneration into the 1Q amplifier circuit to reduce gain. Decreasing 2P resistance increases gain.

Rectifiers 20 and 21D provide a positive bias (about 1.2 volts) to allow the voltage applied to the firing reactor control windings to go to zero when transistor 3Q is full on.

Current Transformer Forcing Unit

The current transformer forcing unit switches in when associated voltage regulator power source fails, allowing its C.T. to supply excitation to the exciter field and thus sustain generator short circuit current as required for selective relay tripping.

Since the voltage regulator output and C.T. forcing unit output are in series, either can supply machine excitation. See Figure 5.



C.T. Forcing Unit Operation

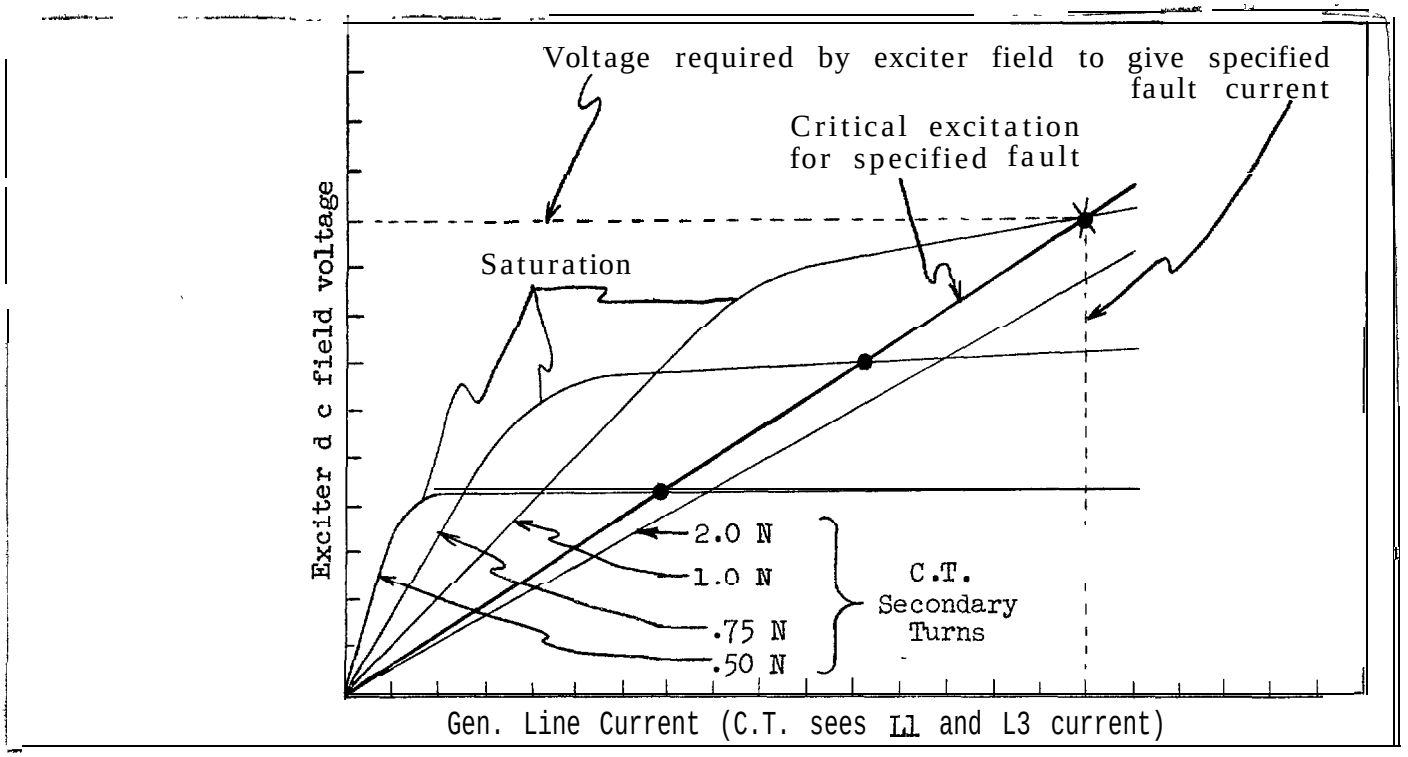
Figure 5

When regulator is operating normally, the CT forcing unit output rectifier bridge (2, 3, 4, and 5D) is shorted by contact 1CR. The voltage regulator, therefore, furnishes voltage to the exciter field normally and the CT current circulates through the shorted bridge.

When the voltage regulator power source fails, or drops below approximately 1/3 normal, relay 1CR will drop out and allow the CT to supply DC current to the exciter field to sustain excitation to the machine.

Since there is no control during the current forcing condition, if the CT output current is too small the system will collapse and not sustain short circuit excitation. If the CT output current is larger than a critical value, the excitation, and short circuit current, will increase until some part of the system saturates.

The forcing C.T. is designed to saturate at an output voltage level as required by thegenerator field for maintaining a specified short circuit current out of the generator. Increasing the C.T. secondary turns increases the voltage level at which the C.T. will saturate; therefore, increases the current sustained into a short circuit. Since increasing the C.T. secondary turns decreases the C.T. secondary current, an excess number of secondary turns will decrease the C.T. secondary current below the critical value and the system will collapse. See Figure 6.



Action of Saturating C.T. on Faults

Figure 6

The setting of the C.T. secondary turns can be established analytically, as shown in Figure 6. For a particular generator, the value of exciter field current can be established for sustaining a specified line current into a specified fault. (Usually a three phase line-to-line fault, the worst condition, is specified). Knowing exciter field current and field resistance,. the required exciter field voltage can be established and the critical excitation line drawn for a specified fault. (See Figure 6).

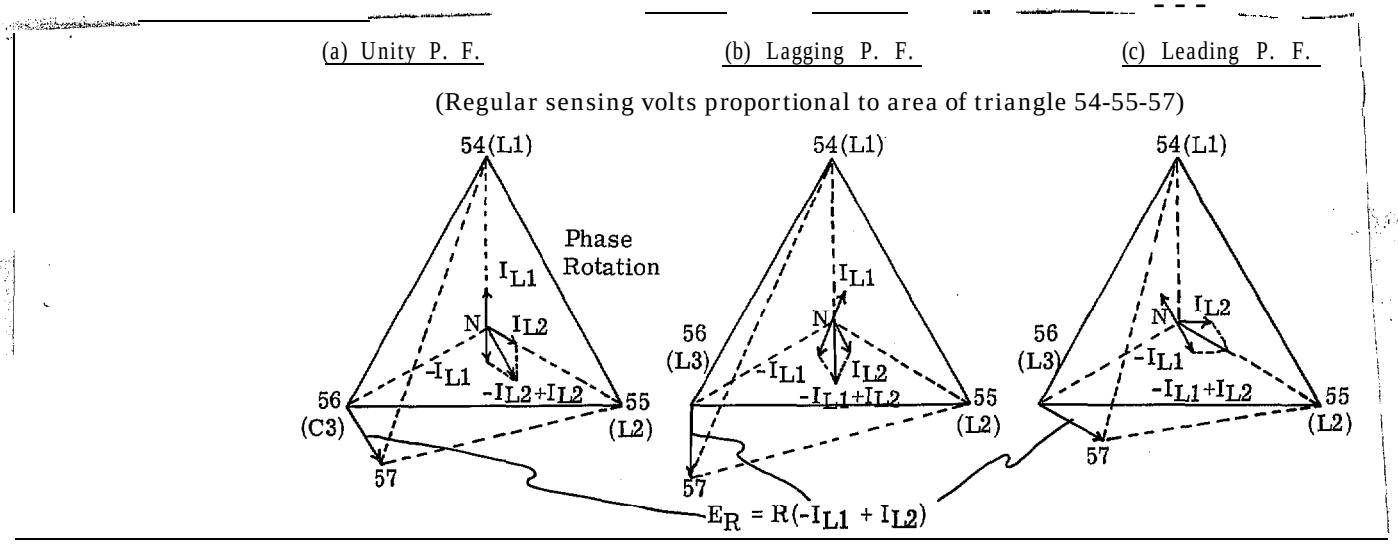
The maximum C.T. turns ratio can be established for the respective line current and exciter field requirements. If the C.T. turns ratio were greater, the C.T. would deliver less than the required excitation, and the system would collapse. The C.T. turns ratio used must be less than the critical value, so that the system will tend to overexcite. The C.T. is then designed to saturate at a voltage level that will provide the proper excitation.

Varying the C.T. secondary turns also varies its voltage saturation level, thus the choice of C.T. turns provides an adjustment. A proper turns ratio can be established as shown in Figure 6 by picking the C.T. turns curve that crosses the critical excitation line nearest the required excitation point X.

The C.T. fault current is extremely high at the first instant a fault occurs and a resistor (1R) is used as a voltage suppressor and limits the excess voltage supplied by the C.T. This protects the rectifier bridges from excess voltage.

"Droop" Circuit for Paralleling Generators

A "Droop" circuit (or line drop compensator) may be connected in series with one leg of the AC regulator sensing circuit. The droop circuit consists of standard 5 amp CT's loaded with a rheostat to deliver an AC voltage that adds to the voltage going to the regulator sensing circuit. This added voltage may cause the sensing circuit voltage to increase or decrease, depending upon the phase relationship. See Figure 7. Where the generator is supplying unity power factor current (Figure 7(a)), the voltage across the rheostat will add to the input voltage at an angle such that the sensing circuit voltage and input voltage will be nearly the same. Should reactive current tend to increase in the lagging direction (Figure 7(b)), the regulator sensing circuit will see a higher than normal voltage, indicating over-excitation; thus, the regulator will decrease excitation to lower line voltage, thereby decreasing the lagging reactive current. Should reactive current tend to increase in the leading direction (Figure 7(c)), the regulator sensing circuit will see a lower than normal voltage, indicating under-excitation; thus, the regulator will increase excitation to raise line voltage, thereby decreasing the leading reactive current.



Operation of Paralleling (Droop) Circuit

Figure 7

When connections from the droop adjusting rheostat are reversed, the reverse of the above described action takes place so that the generator terminal voltage is raised as lagging reactive current increases, the amount of compensation depending upon the setting of the C.T. droop adjusting rheostat. The circuit will therefore compensate for line drop due to reactive current.

Under-Frequency Limit

The Under-Frequency Limit is used to protect the exciter field and the regulator output stages during an under-frequency condition.

A voltage proportional to the generator output is applied to the bridge clipping circuit comprised of the four diodes (22, 23, 24, 25D) and the zener reference (4ZD) through a current limiting resistor (22R). The result is an approximate 48 volts P.P. square wave measured between circuits 64 and 60. The frequency of the square wave is that of the generator output. The saturable reactor (3SX) will saturate after so many volt-seconds are accumulated and a pulse will then appear on its secondary (PTS. 9 and 10). This is adjusted to happen at some frequency under the normal running frequency by moving circuit 60 to one of the various taps on the reactor's primary. The pulses which appear on the secondary winding during the under-frequency condition are rectified by the full wave bridge (26, 27, 28, 29D) and averaged by capacitor (15C). The transistor (4Q) is turned on when the averaged voltage becomes greater than approximately one volt. When the transistor is turned on, it reduces the voltage across the reference to a low value (1.5 volts $\pm$ .5 volts). This requires the regulator to phase back and reduce its output voltage. The resistor (26R) and rectifier (30D) are used to bias the transistor to insure proper turn off when the under-frequency conditions does not exist.

MAINTENANCE

Since the regulator contains no moving parts, little maintenance should be required. Periodic checks should consist of checking the voltage level and regulation. The regulator should be cleaned with a blower as required to prevent an accumulation of dust and dirt.

TROUBLESHOOTING

The following chart may be helpful for troubleshooting and locating faulty components; however, a thorough study of the "Principles of Operation" will be the greatest aid to trouble-shooting.

TROUBLESHOOTING CHART

TROUBLE	PROBABLE CAUSE	CHECK
1. Generator Voltage will not build up	1. No power to generator field	a. Connections to regulator b. Connections between regulator and generator c. 1D d. SCR

TROUBLESHOOTING CHART
(Cont.)

TROUBLE	PROBABLE CAUSE	CAUSE
I. Generator Voltage will not build up (Cont.)	2, No residual voltage out of generator	a. Reverse generator field connections b. Try flashing field with battery (+ to DC+, - to DC-)
II. Generator voltage goes to ceiling	1. No feedback voltage signal	a. "Droop" circuit connection b. 1P connections c. 2ZD, 3ZD
	2. No control of SCR	a. SCR b. SX
III. Poor voltage regulator	1. Loss in system gain	a. "Gain" adjust per OPERATION b. Regulator output to generator field
	2. Distorted input to sensing circuit	a. For unbalanced 3 phase loads b. Generator wave shape
IV. Instability	1. Excess transient system gain	a. "Gain" adjust per OPERATION b. Generator field resistance too low? c. "Stabilizing" per OPERATION
V. Slow system response	1. Low transient system gain	a. "Stabilizing adjustments per OPERATION b. Loss in system gain Check Step III
VI. Paralleled generators will not divide reactive KVA	1. Droop circuit	a. Try reversing connections to D1 and D2 b. Increase droop circuit C.T. loading resistance c. Check generator line phase sequence

TROUBLESHOOTING CHART
(Cont.)

TROUBLE	PROBABLE CAUSE	CAUSE
VII. Sustained short circuit current too low	1. C.T. not deliverying enough voltage	a. Increase C.T. secondary turns
	2. Regulator output low	a. Rectifier 2, 3, 4, 5D
VIII. Sustained short circuit current too high	1. C.T. deliverying too much voltage	a. Decrease C.T. secondary turns
	2. Other than a 3 phase fault	a. C.T. unit and C.T. design is for the worst faults, a 3 phase fault. For single phase faults short circuit 'currents will be higher
IX. System collapses under short circuits	1. 1CR net dropping out	a. Operation of 1CR when voltage is removed
	2. C.T. secondary current too low	a. Decrease C.T. secondary turns

RENEWAL PARTS

Should a component fail, a replacement part can be ordered from the nearest sales office of General Electric Company. When ordering renewal parts, specify the quantity required, give the catalog number and describe the required parts in detail. In addition, give the 3S model number and complete nameplate rating of the equipment. Principal Renewal Parts List is furnished with each particular model. (See dwg, number indicated on nameplates.)

Drawings

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44C321618

44C321640

44C321656

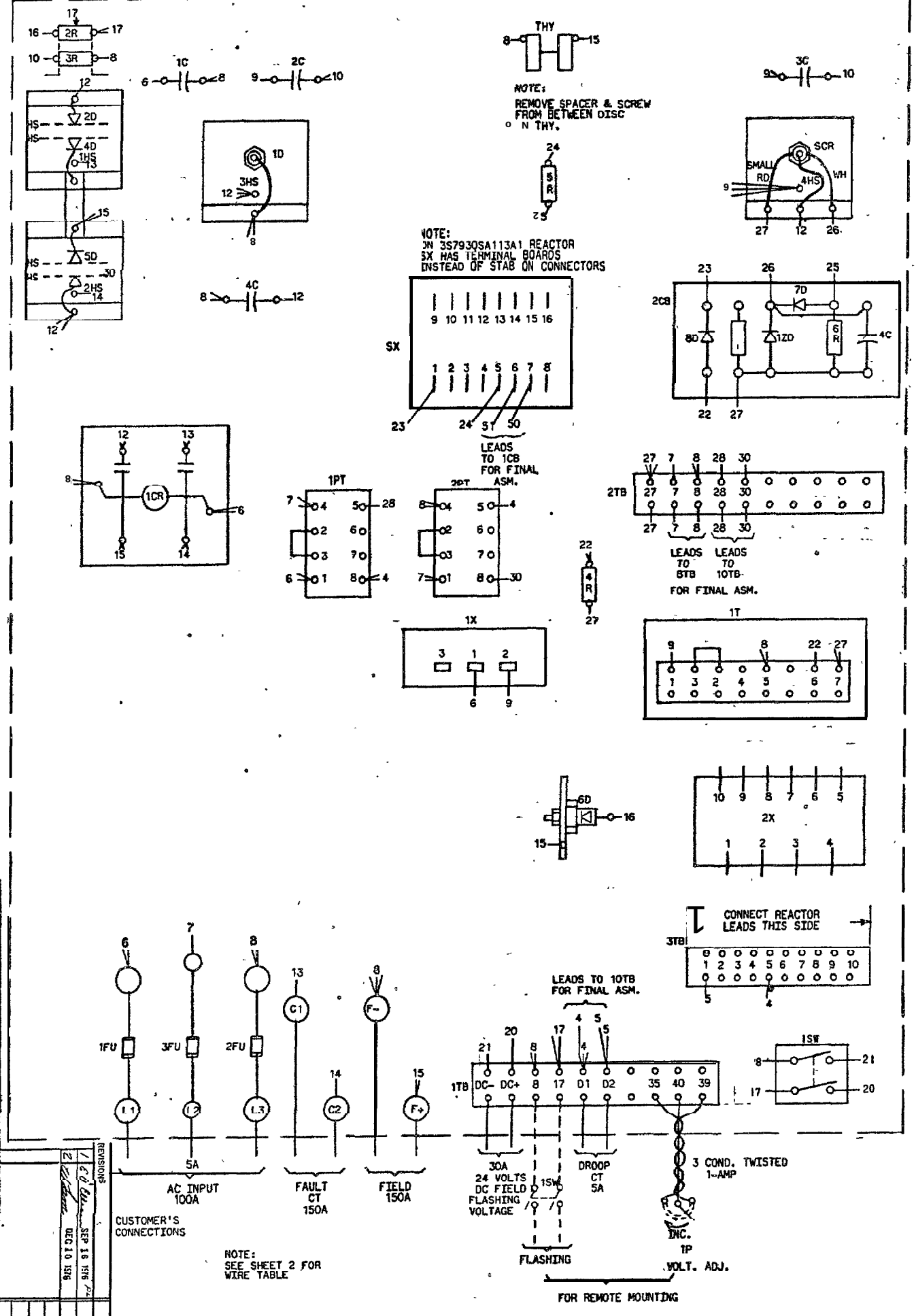
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Figure 8.



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APPROVALS

DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA.

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REVISIONS

2	[illegible]	SEP 18 1978	
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PRINTS TO 7/11

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GENERAL ELECTRIC

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MARINE STATIC EXCITER REGULATOR
CONN. DIAG.

FIRST MADE FOR 3S7930SA112A1, 113A1

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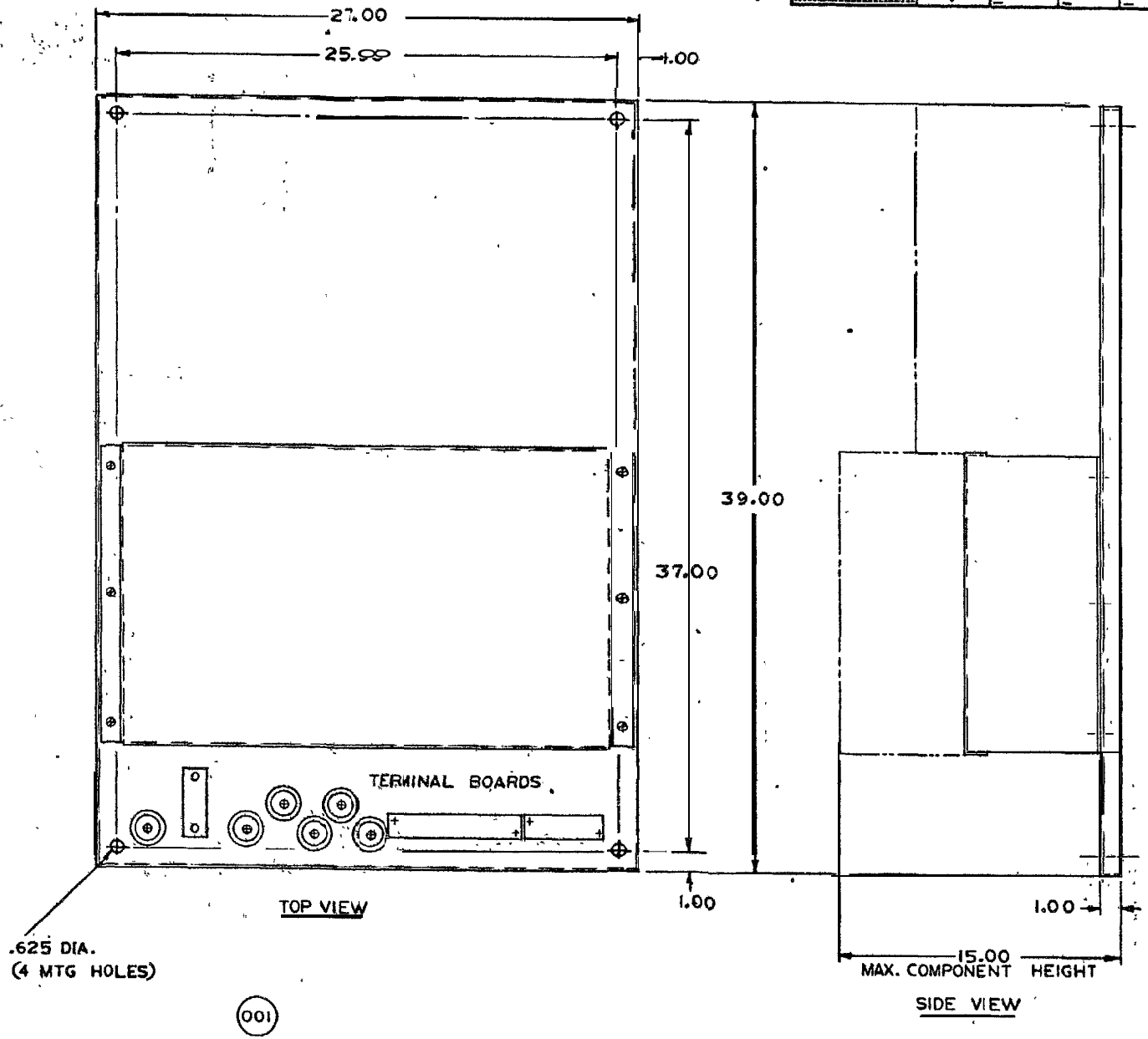
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GENERAL ELECTRIC

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

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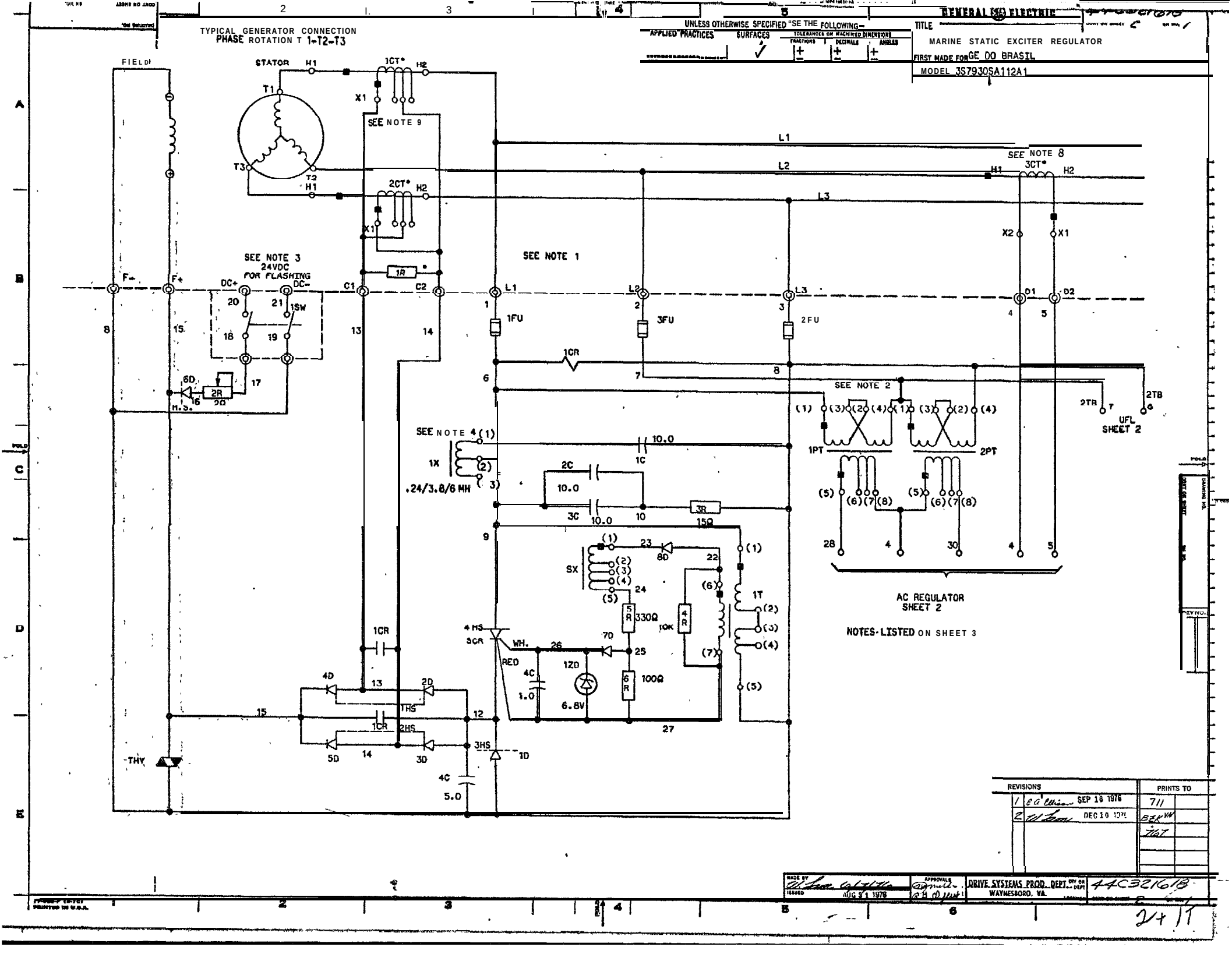
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FIRST MADE FOR 3S7930SA112A1, SA113A1



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GENERAL ELECTRIC

TITLE
MARINE STATIC EXCITER REGULATOR
FIRST MADE FOR GE DO BRASIL
MODEL 3S7930SA112A1

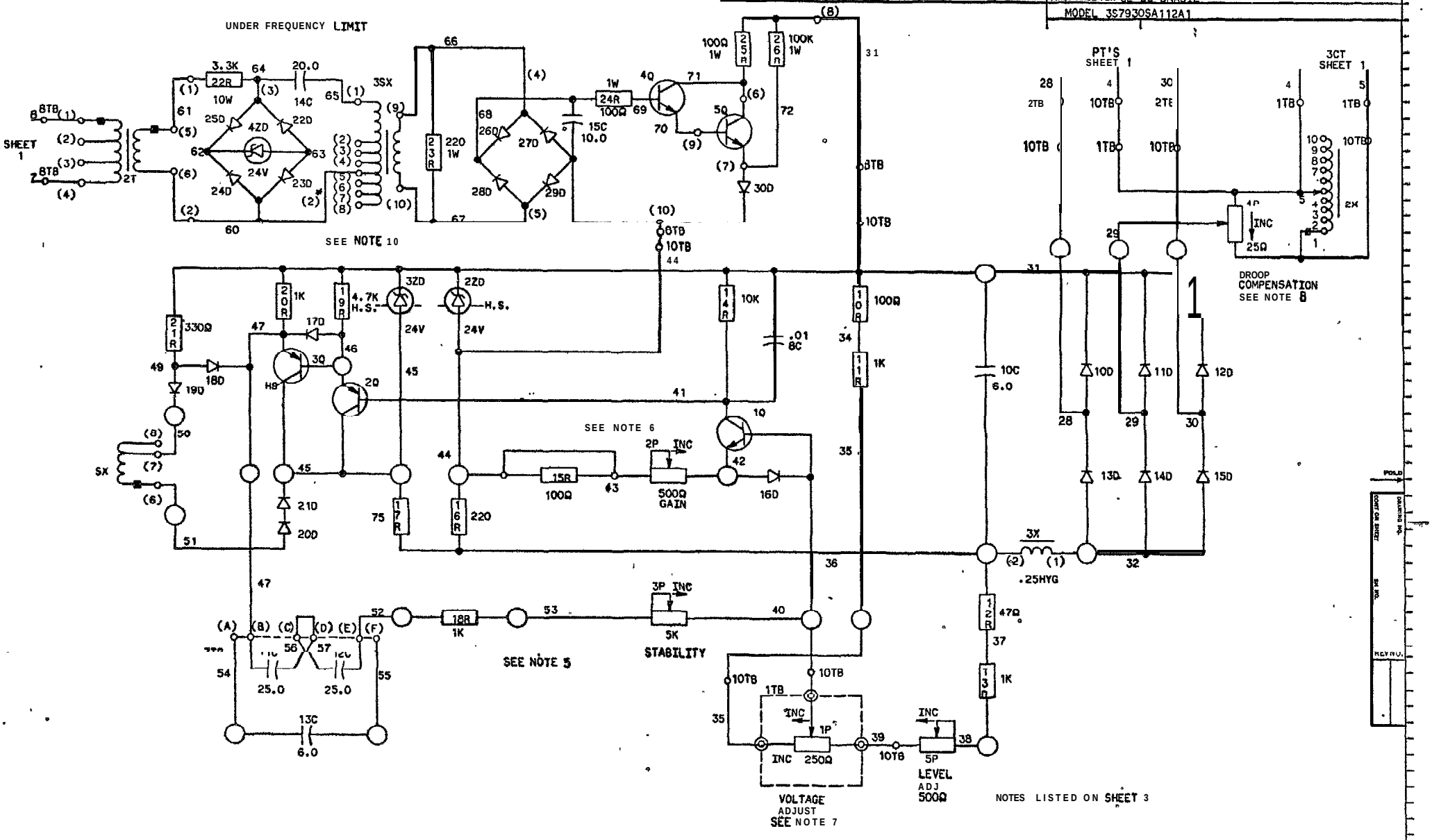
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2	W. Lewis	DEC 10 1976	B2KVA
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APPROVALS Raymond R. H. Wright
DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA.
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TITLE
MARINE STATIC EXCITER REGULATOR
FIRST MADE FOR GE DO BRASIL
MODEL 3S7930SA112A1



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TITLE: MARINE STATIC EXCITER REGULATOR

FIRST MADE FOR GE DO BRASIL

MODEL 3S7930SA112A1

NOTES:

1. STATIC EXCITER IS DESIGNED FOR 450 VOLT 1 PHASE INPUT TO POWER CIRCUIT AT L1 AND L3. FOR OTHER GENERATOR VOLTAGES, STEP DOWN TRANSFORMERS MUST BE USED. TRANSFORMER KVA SHOULD BE APPROX. 2X THE MAX. KW EXCITATION REQUIRED BY THE GENERATOR. PHASE SEQUENCE MUST BE L1-L2-L3.
2. REGULATOR SENSING CIRCUIT IS DESIGNED FOR 440 TO 480 VOLT OR 220 TO 240 VOLT 3 PHASE INPUT. FOR OTHER GENERATOR INPUT VOLTAGES, STEP DOWN POTENTIAL TRANSFORMERS MUST BE USED. INPUT TRANSFORMERS SHOULD BE 100 VOLT-AMPS EACH-MINIMUM.
   A. IF INPUT TO L1, L2, AND L3 IS IN 440 TO 450 VOLT RANGE, JUMPER 2 TO 3 ON 1PT AND 2PT.
   C. IF RANGE OF VOLTAGE ADJ. (1P) IS TOO HIGH, MOVE WIRES 4 AND 30 TO TERM. 8 ON 1T AND 2T.
   SF RANGE OF VOLTAGE ADJUST (1P) IS TOO LOW. MOVE WIRES, AND 30 TO TERM. 6 ON IT AND 2T.
   D. PHASE SEQUENCE MUST BE L1-L2-L3.
3. CONNECT AN EXTERNAL SOURCE OF D.C. TO DC+ AND DC- TO FLASH GENERATOR ON START-UP. FLASHING MAY BE INITIATED BY PUSH-BUTTON OR AUTOMATIC RELAYING BUT MUST NEVER BE CONNECTED IF GENERATOR IS IDLE. RESISTOR 2R IS USED TO ISOLATE FLASHING POWER SUPPLY AND LIMIT FLASHING CURRENT. ADJUST FOR MINIMUM NECESSARY FLASHING AMPS USING JUMPER ON 2R. FLASHING SWITCH, 1SW MAY BE REMOVED FROM EXCITER FOR REMOTE MOUNTING.
4. FILTER CONSISTING OF 1X AN0 3R, IS TO SUPPRESS DISTORTION OF GENERATOR WAVE SHAPE CAUSED BY STATIC EXCITER. FILTER IS SET FOR NOMINAL EFFECT AS SHIPPED, BUT MAY BE CHANGED AS REQUIRED. INDUCTANCE OF 1X IS: 2 TO 3-.24MH., 1 TO 2-3.8MH., 1 TO 3-6.0MH. USE AS LITTLE INDUCTANCE AS POSSIBLE.
5. FOR OPTIMUM TRANSIENT RESPONSE VARY 3P AN0 CAPACITY BETWEEN CIRCUITS 47 AND 52. STABILITY ADJUST 3P ACTS AS A FINE AOJUSTMENT AND CAPACITOR COMBINATION ACTS AS A COARSE OR RANGE ADJUSTMENT.
   FOR 50 MFD JUMPER B TO C AND D TO E ON 7TB.
   FOR 20 MFD JUMPER B TO C ON 7TB.
   FOR 12.5 MFD JUMPER 0 TO C ON 7TB.
   FOR 6 MFD JUMPER A TO B AND E TO F ON 7TB.
6. INCREASING GAIN WILL IMPROVE STEADY STATE REGULATION. GAIN SHOULD BE SET AS HIGH. AS POSSIBLE, SO LONG AS SYSTEM IS NO? UNSTABLE. REMOVE JUMPER ON 15R TO LIMIT MAXIMUM GAIN.
7. THE "VOLTAGE ADJUST" 1P IN A.C. REGULATOR, MAY BE REMOVED AND REMOTELY MOUNTED IF REQUIRED. ADJUST 5P TO CHANGE RANGE OR VOLTAGE ADJUST 1P.
8. CONNECT DROOP CT AS SHOWN FOR DIVISION OR REACTIVE CURRENT WHEN PARALLELING GENERATORS. WITH GENERATOR CONNECTIONS AS SHOWN TO L1-L2-L3, CT MUST BE IN LINE L2 WITH POLARITY AS SHOWN. INCREASE 4P TO INCREASE REACTIVE COMPENSATION. FOR SINGLE GENERATOR OPERATION WITHOUT DROOP, JUMPER D1 TO D2.
9. 1CT AND 2CT SECONDARY CONNECTIONS MAY BE VARIED TO OBTAIN VARIOUS AMOUNTS OR FORCING. INCREASING SECONDARY TURNS INCREASES FORCING. SEE TABLE 1 FOR TURNS RATIO AND TABLE 2 FOR SUGGESTED GENERATOR RATING.
10. ADJUST LEAR 60 ON REACTOR TAPS 2 TO 8 TO POSITION WHERE AC VOLTAGE BEGINS TO DECREASE. THEN MOVE TO NEXT LOWER TAP. LET GENERATOR COAST TO STOP AND OBSERVE FIELD CURRENT, FIELD CURRENT SHOULD NOT INCREASE MORE THAN 20 TO 30%. OBSERVE TRANSIENT RESPONSE BY APPLYING AND REJECTING LOAD. UNDERFREQUENCY LIMIT SHOULD BE SET ON TAP WHERE TRANSIENT RESPONSE IS NOT AFFECTED.

LEGEND

- CIRCUIT TERMINAL POINT
- CUSTOMER CONNECTION TO EXCITER
- REGULATOR BOARD CONNECTION
- * FURNISHED SEPARATE FOR CUSTOMER MOUNTING.

TABLE 1

CT	TAPS	RATIO	VOLTS
44B376814-G01	1-2	10	71
	1-3	14	100
	1-4	18	128
44B374295-G01	1-2	23	99
	1-3	27	116
	1-4	31	133

TABLE 2

GEN. KW	1CT, 2CT	TAPS	1R	3CT	3CT RATIO
310	44B376814-G01	1-3	274A906G7	822X86	800:5
375	44B374295-G01	1-2	274A906G9	822X87	1000:5
480	44B374295-G01	1-3	274A906G9	822X87	1000:5

DRAWING NO. 44C321618 — CONT ON SHEET FILL. SH NO. 3 — REV NO

REVISIONS

PRINTS TO: 711; 82X VM; 76T

MADE BY: [illegible] 6/4/76 — ISSUED: AUG 31 1976 — APPROVALS: [illegible] — DRIVE SYSTEMS PROD. DEPT. DIV OR DEPT — WAYNESBORO, VA. LOCATION — 44C321618 — CONT ON SHEET FILL. SH NO. 3

GENERAL ELECTRIC

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES OR MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
MARINE STATIC EXCITER REGULATOR
FIRST MADE FOR GE DO BRASIL
MODEL 3S7930SA113A1

44C321640 CONT ON SHEET 2 DR NO. 1

TYPICAL GENERATOR CONNECTION
PHASE ROTATION T1-T2-T3

FIELD
STATOR H1 1CT* H2
T1 X1
T3 T2
SEE NOTE 9
H1 2CT* H2
X1
SEE NOTE 1
L1
L2
L3
SEE NOTE 8
3CT* H1 H2
X2 X1
SEE NOTE 3
24VDC
FOR FLASHING
DC+ DC- C1 C2
F- F+
1R
20 21 1SW
10 19
6D 2R 2Ω 16 17
L1 L2 L3 D1 D2
1FU 3FU 2FU
1CR
SEE NOTE 2
(1) (3) (2) (4) (1) (3) (2) (4)
1PT 2PT
2TB 7 2TB 8
UFL
SHEET 2
SEE NOTE 4 (1)
1X
(2)
(3)
.24/3.8/6 MH
10.0 1C
2C 10.0
3C 10.0 10 3R 15Ω
SX
(1) (2) (3) (4) (5)
23 6D 22 24
5R 330Ω
4R 10K
1T (1) (2) (3) (4) (5) (6) (7)
HS SCR
26 7D 25
1ZD 6.8V
6R 100Ω
4C 1.0
1CR
4D 2D
13
HS
15 12
1CR
HS
5D 14 30
THY
4C 5.0
3HS 1D
27
AC REGULATOR
SHEET 2
NOTES LISTED ON SHEET 3

REVISIONS
1 [illegible] SEP 16 1976
2 [illegible] DEC 10 1976

PRINTS TO
711
BEK
MoT

MADE BY J.R. Kuster 7-1-76 ISSUED AUG 3 1 1976
APPROVALS
DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA.
44C321640 CONT ON SHEET 2 DR NO. 1

2 + 17

GENERAL ELECTRIC | 44C321640 CONT ON SHEET 3 DR NO. 2

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
MARINE STATIC EXCITER REGULATOR
FIRST MADE FOR GE DO BRASIL
MODEL 3S7930SA113A1

UNDER FREQUENCY LIMIT

SHEET 1 | 8TB | 7TB | 6TB | 21T

3.3K 22R 10W | 20.0 14C | 35X | 25D 4ZD 22D | 24V | 24D 23D | 60 | SEE NOTE 10

66 | 220 1W 23R | 68 26D 27D 28D 29D | 67 | 15C 10.0 | 1W 24R 100Ω 69 | 40 71 50 70 | 100Ω 1W 25R | 100K 1W 26R | 72 | 30D

(B) | 31 | 8TB | 10TB | 44

PT'S SHEET 1 | 3CT SHEET 1 | 2TB | 10TB | 1TB | 4P INC 25Ω | BROOP COMPENSATION SEE NOTE 8 | 2X

20R 1K | 19R 4.7K H.S. | 3ZD | 2ZD H.S. | 24V | 24V | 14R 10K | 10R 100Ω | 11R 1K | 8C .01 | 10C 6.0

21R 330Ω | 47 | 17D | 49 | 16D | 19D | 30 HS | 46 | 2Q | 45 | 41 | 1Q | 34 | 35

SEE NOTE 6 | 2P INC 500Ω GAIN | 15R 100Ω | 43 | 42 | 16D | 44

5X | 50 | 51 | 21D | 20D | 17R 75 | 16R 220

10D 11D 12D | 13D 14D 15D | 28 29 30 | 32

3X .25HYG | 36

47 | 52 | 18R 1K | 53 | 3P INC 5K STABILITY | 40 | 12R 47Ω | 317 | 13R 1K

7TB | (A) (B) (C) (D) (E) (F) | 56 57 | 11C 25.0 | 12C 25.0 | 54 | 55 | 13C 6.0 | SEE NOTE 5

10TB | 10TB | 35 | INC 1P 25Ω | 39 | 10TB | INC 2P LEVEL ADJ. 5000 | 38 | VOLTAGE ADJUST SEE NOTE 7

NOTES LISTED ON SHEET 3

REVISIONS
1 | E.A. Eu | SEP 16 1976

PRINTS TO
711
GZK VM
TGT

MADE BY R.R. Kuster 7-1-76 | ISSUED AUG 31 1978 | APPROVED Eugene L. R.H. Spurr
DRIVE SYSTEMS PROD. DEPT. DIV OR DEPT. WAYNESBORO, VA. LOCATION
44C321640 CONT ON SHEET 3 DR NO. 2

PF-060-F (8-75) PRINTED IN U.S.A.

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE MARINE STATIC EXCITER REGULATOR
FIRST MADE FOR GE DO BRASIL
MODEL 3S7930SA113A1

NOTES:

1. STATIC EXCITER IS DESIGNED FOR 450 VOLT 1 PHASE INPUT TO POWER CIRCUIT AT LI AND L3. FOR OTHER GENERATOR VOLTAGES, STEP DOWN TRANSFORMERS MUST BE USED. TRANSFORMER KVA SHOULD BE APPAOX. 2X THE MAX. KW EXCITATION REQUIRED BY THE GENERATOR. PHASE SEQUENCE MUST BE L1-L2-L3.

2. REGULATOR SENSING CIRCUIT IS DESIGNED FOR 440 TO 460 VOLT OR 220 TO 240 VOLT 3 PHASE INPUT. FOR OTHER GENERATOR INPUT VOLTAGES, STEP DOWN POTENTIAL TRANSFORMERS MUST BE USEO. INPUT TRANSFORMERS SHOULD BE 100 VOLT-AMPS EACH-MINIMUM.
   A. IF INPUT TO L1, L2, AND L3 IS IN 440 TO 450 VOLT RANGE, JUMPER 2 TO 3 ON 1PT AND 2PT.
   G. IF RANGE OF VOLTAGE AOJ. (1P) IS TOO HIGH, MOVE WIRES 4 AND 30 TO TERM. 8 ON 1T AND 2T.
   IF RANGE OF VOLTAGE ADJUST (1P) IS TOO LOW, MOVE WIRES 4 AND 30 TO TERM. 6 ON 1T AND 2T,
   D. PHASE' SEQUENCE MUST BE L1-L2-L3.

3. CONNECT AN EXTERNAL SOURCE OF D.C. TO DC+ AND OC- TO FLASH GENERATOR ON START-UP. FLASHING MAY BE INITIATED BY PUSH-BUTTON OR AUTOMATIC RELAYING BUT MUST NEVER BE CONNECTED IF GENERATOR Is IDLE. RESISTOR 2R IS USED TO ISOLATE FLASHING POWER SUPPLY AND LIMIT FLASHING CURRENT. ADJUST FOR MINIMUM NECESSARY FLASHING AMPS USING JUMPER ON 2R. FLASHING SWITCH, 1SW MAY BE REMOVED FROM EXCITER FDA REMOTE MOUNTING.

4. FILTER CONSISTING Of 1X AND 3R IS TO SUPPRESS DISTORTION OF GENERATOR WAVE SHAPC CAUSEO BY STATIC EXCITER. FILTER IS SET FOR NOMINAL EFFECT AS SHIPPED. BUT MAY BE CHANGED AS REQUIRED. INDUCTANCE OF 1X IS: 2 TO 3-.24MH., 1 TO 2-3.8MH., 1 TO 3-6.OMH. USE AS LITTLE INDUCTANCE AS POSSIBLE.

5. FOR OPTIMUM TRANSIENT RESPONSE VARY 3P AND CAPACITY BETWEEN CIRCUITS 47 AND 52. STABILITY ADJUST 3P ACTS AS A FINE ADJUSTMENT AND CAPACITOR COMBINATION ACTS AS A COARSE OR RANGE AOJUSTMENT,
   FOR 50 MFD JUMPER B TO C AND D TO E ON 7TB.
   FOR 20 MFD JUMPER B TO C ON 7TB.
   FOR 12.5 MFD JUMPER D TO C ON 7TB.
   FOR 6 MFD JUMPER A TO B AND E TO F ON 7TB.

6. INCREASING GAIN WILL IMPROVE STEADY STATE REGULATION. GAIN SHOULD BE SET AS HIGH AS POSSIBLE, so LONG AS SYSTEM IS NOT UNSTABLE. REMOVE JUMPER ON 15R TO LIMIT MAXIMUM GAIN.

7. THE "VOLTAGE ADJUST" 1P IN A.C. REGULATOR MAY BE REMOVED ANO REMOTELY MOUNTED IF REQUIRED. ADJUST 5P TO CHANGE RANGE OR VOLTAGE ADJUST 1P.

8. CONNECT DROOP CT AS SHOWN FOR DIVISION OR REACTIVE CURRENT WHEN PARALLELING GENERATORS. WITH GENERATOR CONNECTIONS AS SHOWN TO L1-L2-L3, CT MUST BE IN LINE L2 WITH POLARITY AS SHOWN. INCREASE 4P TO INCREASE REACTIVE COMPENSATION. FOR SINGLE GENERATOR OPERATION WITHOUT DROOP. JUMPER D1 TO D2.

9. 1CT AND 2CT SECONDARY CONNECTIONS MAY BE VARIED TO OBTAIN VARIOUS AMOUNTS OR FORCING. INCREASING SECONDARY TURNS INCREASES FORCING. SEE TABLE 1 FOR TURNS RATIO AND TABLE 2 FOR SUGGESTED GENERATOR RATING.

10. ADJUST LEAD 60 ON REACTOR TAPS 2 TO 8 TO POSITION WHERE AC VOLTACE BEGINS TO DECREASE, THEN MOVE TO NEXT LOWER TAP. LET GENERATOR COAST To STOP ANO OBSERVE FIELD CURRENT. FIELD CURRENT SHOULD NOT INCREASE MORE THAN 20 TO 30%. OBSERVE TRANSIENT RESPONSE BY APPLYING AND REJECTING LOAD. UNDERFREQUENCY LIMIT SHOULD BE SET ON TAP WHERE TRANSIENT RESPONSE IS NOT AFFECTED.

LEGEND

- CIRCUIT TERMINAL POINT
- CUSTOMER CONNECTION TO EXCITER
- REGULATOR BOARD CONNECTION
- * FURNISHED SEPARATE FOR CUSTOMER MOUNTING.

TABLE 1

CT	TAPS	RATIO	VOLTS
44B376014-G01	1-2	10	100
	1-3	14	
	1-4	18	
44B374295-G01	1-2	23	116
	1-3	27	
	1-4	31	

TABLE 2

GEN. KW.	1CT, 2CT	TAPS	1R	3CT	3CT RATIO
520	44B374295-G01	1-3	274A9066G9	822X88	1200:5
550	44B374295-G01	1-4	274A9066G9	822X88	1200:5

FOLD

COUNT ON SHEET | DRAWING NO. | SH NO.

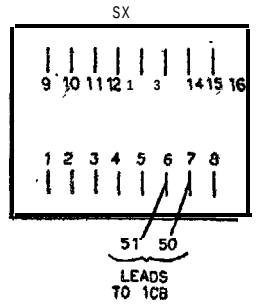
REV NO.

REVISIONS | PRINTS TO 7/1 BEK W 1eL

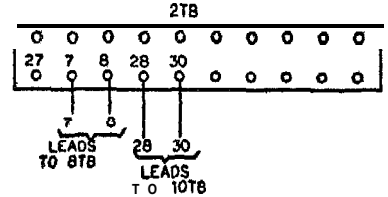
MADE BY J.R. Ristic 7-7-76 ISSUED AUG 3 1 1976
APPROVED E.A. Dinelli
DRIVE SYSTEMS PROD. DEPT. DIV OR DEPT. WAYNESBORO, VA. LOCATION
44C321640 CONT ON SHEET FWL. SH NO. 3

TITLE
MARINE STATIC EXCITER REGULATOR
FIRST MADE FOR 3S7930SA112A1, 113A1
CONN - INTERCONN. DIAG.

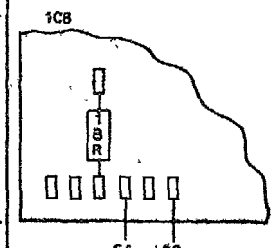
3S7930SA112A1
CONN. DIAG. 44C321656 SH. 1

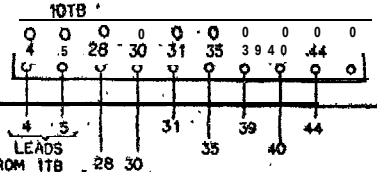


NOTE:
REACTOR SX ON 3S7930SA113A1 HAS TERMINAL BOARDS INSTEAD OF STAB. ON TERMINALS

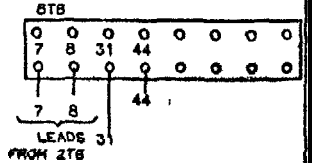


3S7932MA196C2
CONN. DIAG. UC321657

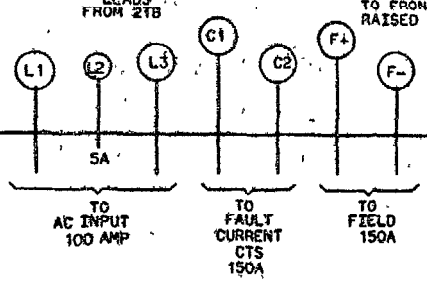


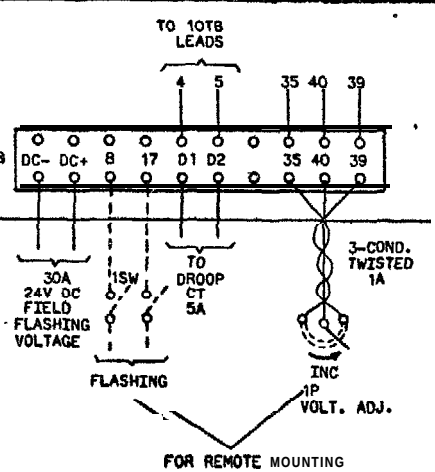


3S7932YA114A6
CONN. DIAG. 44C321658



CLAMP LEADS TO FRONT OF RAISED PANEL





CUSTOMER'S CONNECTIONS

WIRE TABLE FOR 3S7930SA112A1, 113 SH. 1

CIR. NO.	WIRE CODE	SEQUENCE of CONNECTIONS
4	C	2PT, 1PT, 1TB, 3TB
5	C	3TB, 1TB
6	H	1X, 1FU
6	C	1C, 1PT, 1CR, 1FU
7	C	2TB, 2PT, 1PT, 3FU
8	H	THY, 1D, 2FU, F-
8	C	3R, 1C, 4C, 2PT, 2TB, 1T, 1CR, F-, 1TB, 1SW
9	H	4HS, 1X
9	C	2C, 3C, 4HS, 1T
10	C	3R, 2C, 3C
12	H	2D, 3D, 1CR, 3HS, SCR
12	C	3HS, 4C
13	H	1HS, 1CR.C1
14	H	2HS, 1CR.C2
15	H	THY, 5D, 1CR, F+
15	E	6D, F+
16	E	2R, 6D
17	E	2R, 2R, 1TB, 1SW
20	E	1SW, 1TB
21	E	1SW, 1TB
22	B	2CB#, 4R, 1T
23	B	2CB#, SX# •
24	B	5R, SX# •
25	B	5R, 2CB#
26	B	SCR, 2CB#
27	B	SCR, 2TB, 2CB#
27	B	2TB, 1T, 4R
28	C	1PT, 2TB
30	C	2TB, 2PT

• ☒ R☒ 3S7930SA113A1 THESE ☒☒ R☒ SX CONNECT TO A TERMINAL BOARD.

REACTOR 2X

LEAD NO.	CIR. NO.	CONNECT TO
1	5	3TB-1
2		3TB-2
3		3TB-3
4		3TB-4
5	4	3TB-5
6		3TB-6
7		3TB-7
8		3TB-8
9		3TB-9
10		3TB-10

INTER CONNECTION TABLE

CIR. NO.	WIRE CODE	CONNECT TO
31	C	10TB, 8TB
35	C	10TB, 1TB
39	C	10TB, 1TB
40	C	10TB, 1TB
44	C	10TB, 8TB

LEAD TABLE

LEAD NO.	WIRE CODE	LENGTH	CONNECT TO
4	C	30"	1TB-4, 10TB
5	C	30"	1TB-5, 10TB
7	C	40"	2TB-7, 8TB
8	C	40"	2TB-8, CTB
28	C	40"	2TB-28, 10TB
30	C	40"	2TB-30, 10TB
*50	B	30"	SX-7#, 1CB#
*51	B	30"	SX-6#, 1CB#

• RUN THESE WIRES SHORTEST WAY FROM SX TO 1CB. ON 3S7930SA113A1 THESE LEADS ON SX CONNECT TO A TERMINAL, BOARD.

REVISIONS			PRINTS TO
1	C.D. Ellis	SEP 16 1976	711
2	W. Lewis	DEC 10 1976	BZK

MADE BY C. Ellis 8/11/76 ISSUED AUG 8 1975 | APPROVALS | DRIVE SYSTEMS PROD DEPT | 44C321656 CONT ON SHEET FL SH NO. 2

1 of 17

GENERAL ELECTRIC

TITLE
MARINE STATIC EXCITER REGULATOR

FIRST MADE FOR 3S7932MA196C2

CONN. DIAG.

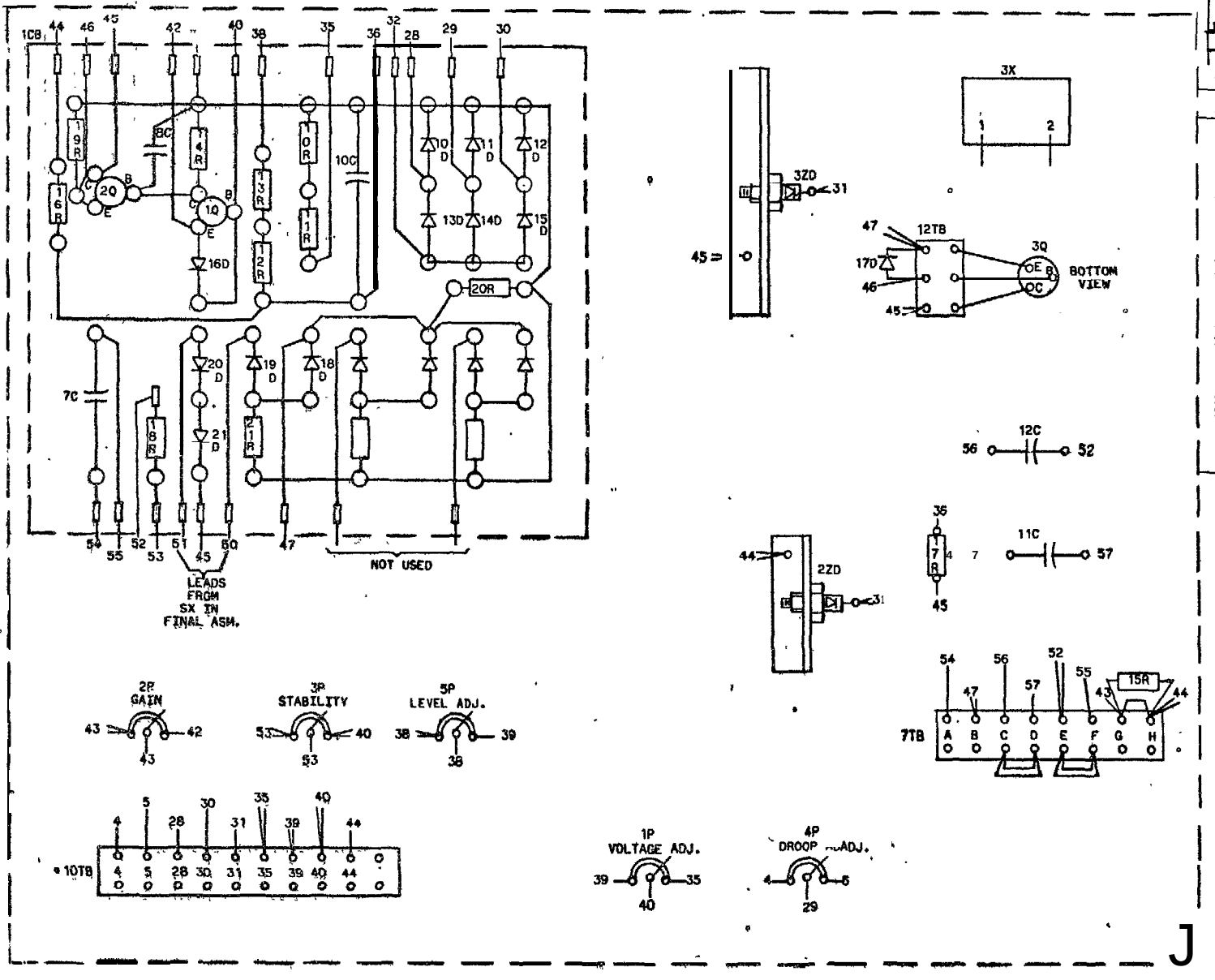
WIRE TABLE FOR 3S7932MA 196C2

CIR. NO.	WIRE CODE	CONNECT TO
4	B	10TB,4P
5	B	10TB,4P
28	B	1CB#,10TB
29	B	1CB#,4P
30	B	1CB#,10TB
31	B	1CB#,3ZD,2ZD,10TB
35	B	1CB#,10TB,1P
36	B	1CB#,17R
38	B	1CB#,5P,5P
39	B	5P,10TB,1P
40	B	1CB#,3P,10TB,1P
42	B	1CB#,2P
43	B	2P,2P,7TB
44	B	1CB#,2ZD,7TB,10TB
45	B	*1CB#,3ZD,12TB,17R,1CB#
46	B	1CB#,12TB
47	B	1CB#,12TB,7TB,11C
52	B	1CB#,7TB,12C
53	B	1CB#,3P,3P
54	B	1CB#,7TB
55	B	1CB#,7TB
56	B	12C,7TB
57	B	11C,7TB

* TO 45 @ TOP OF 1CB

3X REACTOR

LEAD NO.	CIR. NO.	CONNECT TO
1	32	1CB#
2	36	1CB#



J

REVISIONS			PRINTS TO
1	E.A.Ellen	SEP 16 1976	711
2	P.R.Hunt	SEP 21 1976	
3	ADDED JUMPER TO 20E Pedersen	JAN 4 1977	

MADE BY ISSUED AUG 31 1976 — APPROVALS — DRIVE SYSTEMS PROD. DEPT — WAYNESBORO, VA. — LOCATION — 44C321657 — SHT NO. 2 OF SHEETS — SH NO. 1

GENERAL ELECTRIC

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

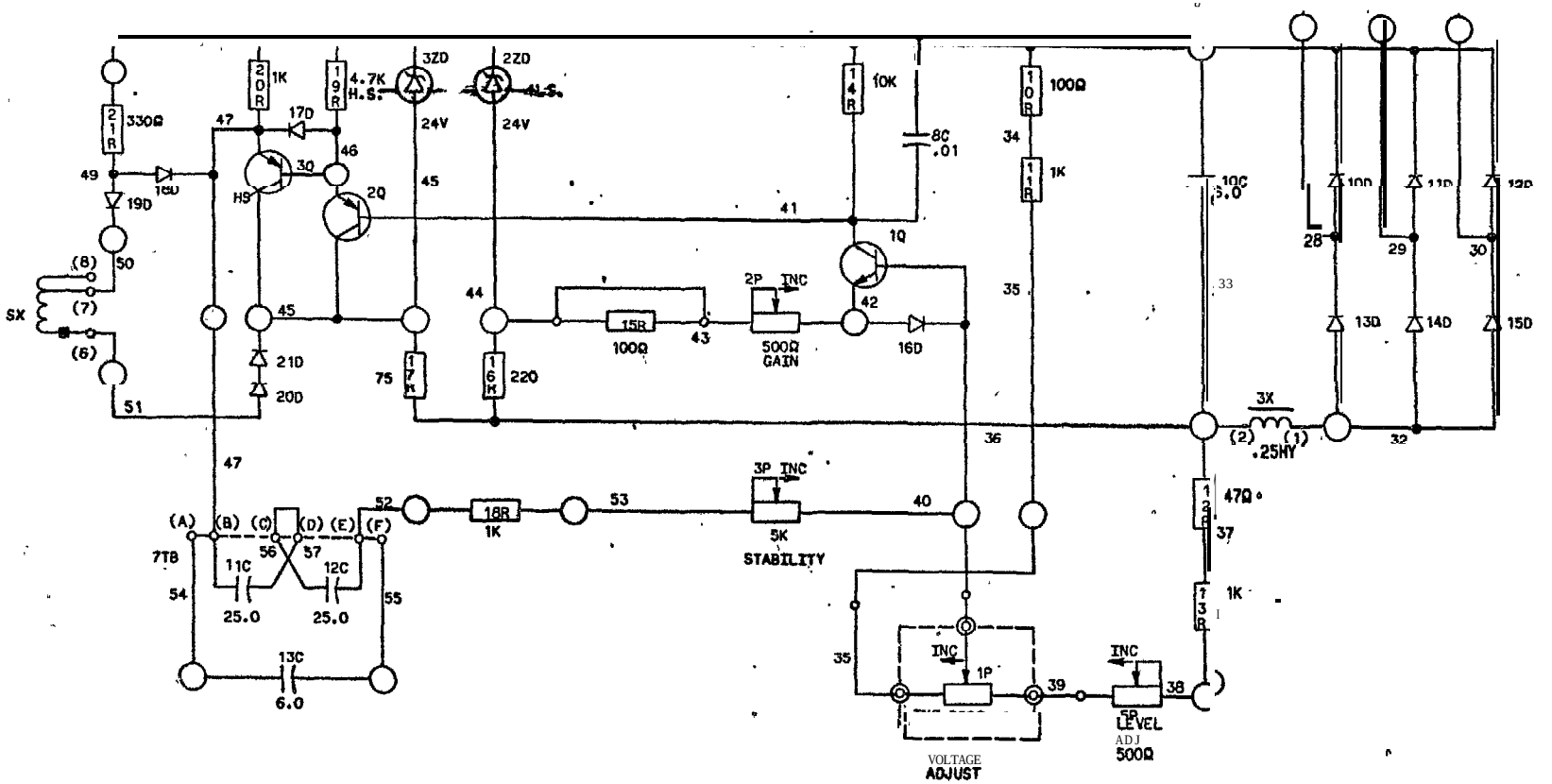
TITLE

MARINE STATIC EXCITER REGULATOR

FIRST MADE FOR GE DO BRASIL

MODEL 3S7932MA196C2 ELEM. DIAG.

44C321657



REVISIONS

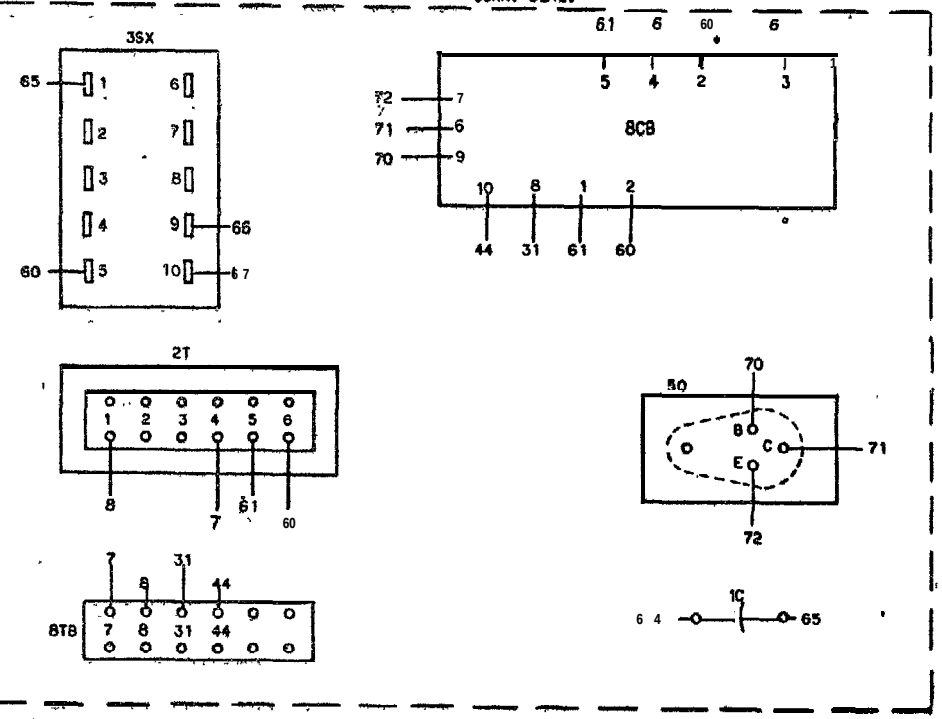
PRINTS TO 711

DRAWN BY	APPROVALS	DRIVE SYSTEMS PROD. DEPT.	44C321657
ISSUED Aug 6, 1976		WAYNESBORO, VA.	

GENERAL ELECTRIC

TITLE
MARINE STATIC EXCITER REGULATOR
FIRST MADE FOR 3S7932YA114A6
ELEM-CONN DIAG.

CONN. DIAG.



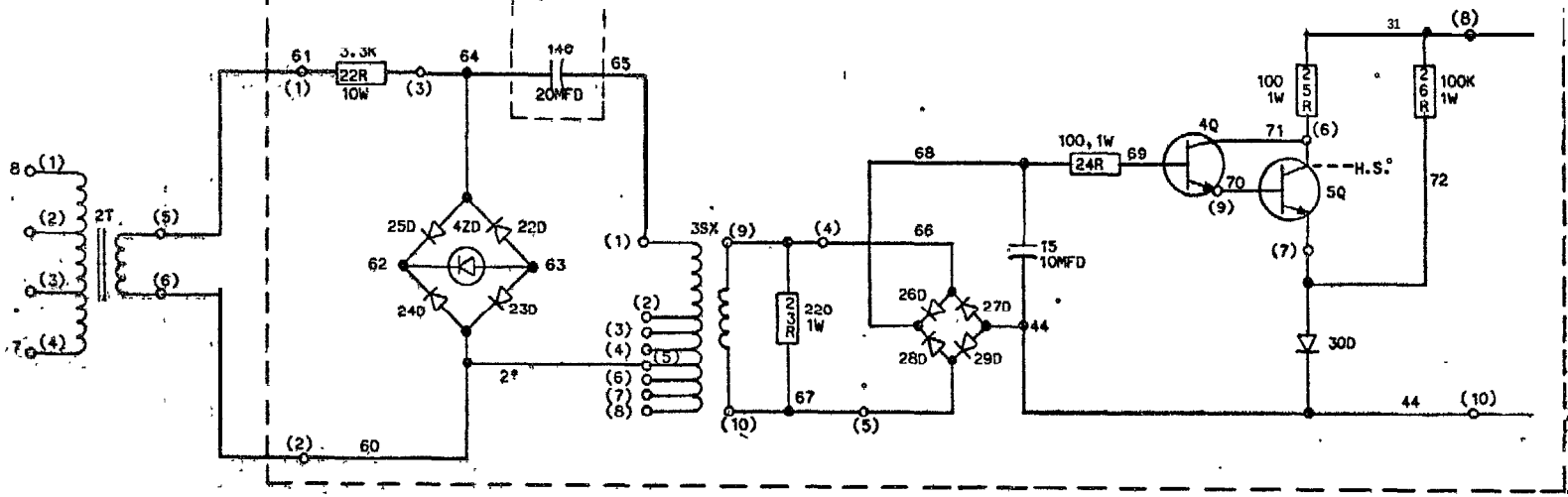
WIRE TABLE

CIR. NO.	WIRE CODE	CONNECT TO
7	B	2T,8TB
8	B	2T,8TB
65	B	3SX-1,14C

8CB

LEAD NO.	CIR. NO.	CONNECT TO
1	61	2T-5
2	60	2T-6
2*	60	3SX-5
3	64	14C
4	66	3SX-9
5	67	3SX-10
6	71	5Q(C)
7	72	5Q(E)
8	31	8TB-31
9	70	5Q(B)
10	44	8TB-44

ELEM. DIAG.



REVISIONS

1	E.A. Ellis	SEP 16 1976

PRINTS TO
711

MADE BY E.A. Ellis 8/12/76
APPROVED
AUG 31 1976
DRIVE SYSTEMS PROD. DEPT
WAYNESBORO, VA
44C321658

GENERAL ELECTRIC

44A376059

REV NO. | TITLE | CONT ON SHEET 2 | SH NO. 1

PRINCIPAL RENEWAL PARTS LIST FOR
3S7930SA112A1

CONT ON SHEET | SH NO. | FIR' | MADE FOR

ITEM NO.	CIRCUIT SYMBOL	QTY.	CATALOG NUMBER	DESCRIPTION OF PART	REVISIO
1	2x	1	44B376804-G01	REACTOR	
2	SX	1	44B219907-G01	REACTOR	
3	1PT,2PT	2	44B311312-G01	TRANSFORMER	
4	SCR	1	44B313954-002	RECTIFIER, 180A, 1300V	
5	1,2,3D	3	44A319818-006	RECTIFIER,250A,1200V	
6	4,5D	2	44A319818-106	RECTIFIER,250A, 1200.V	
7	1T	1	44B334557-G01	TRANSFORMER	
8	1,2,3 C	3	28F5 158FC	CAPACITOR,10MFD,2000V	
9	4C	1	28F5 156FC	CAPACITOR,5MFD,2000V	
10	3R	1	44A333454-D15	RESISTOR,15 OHMS,225W	
11	IX	1	44B214515-G01	REACTOR	
12	1,2FU	2	933B578P17	FUSE 100A, 600V	
13	3FU	1	K977470 1 P2	FUSE 3A;600V	
14	FOR 3FU	1	TC8050	FUSE BLOCK	
15	2R	1	44A333458-C20	RESISTOR,2.0 OHMS,225W	
16	6D	1	IN2 160	RECTIFIER 25A,600V	
17	FOR 6D,4R,5R	3	281A482P4	STANDOFF INSUL.	
18	1CR	1	CR205M004	CONTACTOR	
19	THY	1	9RV6A2	THYRITE	
20	FOR POWER INPUT	8	44A311988-001	INSULATOR	
21	1,2TB	3	22010	TERMINAL BOARD	
22	1SW	1	806 30	SWITCH	
33	4R	1	44A333451-G10	RESISTOR., 10K,25W	
24	5R	1	K86222 12E33	RESISTOR,330 OHMS,10W	
25	FOR 4R	2	44A339057-001	MICA WASHER	
26	FOR 2R,3R	4	44A339057-003	MICA WASHER	
27	AC REGULATOR	1	3S7932MA 19662	AC REGULATOR CONSISTING OF:	
28	2P,5P	2	44A3379 16- 104	POTENTIOMETER,500 OHMS 2W	
29	3P	1	M9729433F50	POTENTIOMETER, .5000 OHMS,2W	
30	1P	1	44A3379 16- 103	POTENTIOMETER 250 OHMS,2W	
31	4P	1	M9729350D25	POTENTIOMETER 25 OHMS, 25W	
32	3x	1	44B211589-G01	D. C. REACTOR	
33	11,12C	2	44A370874-K25	CAPACITOR,25MFD,200VDC	

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MADE BY C.A. Ellison 8/10/76 | APPROVALS | DRIVE SYSTEMS PROD. DEPT. | DIV OR DEPT. | 44A376059

ISSUED AUG 26 1976 | WAYNESBORO, VA.

GENERAL ELECTRIC

44A376059

CONT ON SHEET 3 SH NO. 2

REV NO.

CONT ON SHEET SH NO.

TITLE PRINCIPAL. RENEWAL PARTS LIST FOR 3S7930SA112A1

FIRST MADE FOR

ITEM NO.	CIRCUIT SYMBOL	QTY.	CATALOG NUMBER	DESCRIPTION OF PART
3 4	3Q	1	2N3215	TRANSISTOR
35	2ZD	1	44A337971-017	ZENER DIODE,24V,10W,5%
36	3ZD	1	44A337972-011	ZENER DIODE,24V,50W,5%
37.	FOR 2,3ZD H.S.	4	281A482P2	STANDOFF INSULATOR
38	10TB	1	22010	TERMINAL BOARD
39	7TB	1	22008	TERMINAL BOARD
40	12TB	1	22003	TERMINAL BOARD
41	17R	1	K86222 13D75	RESISTOR 75 OHMS,20W
42	170	1	44B2320 19-004	RECTIFIER 1A,400V
43	15R	1	44A33725 I-E IO	RESISTOR 100 OHMS, 2W
44	REGULATOR BOARI	1*	3S7932MA 19882	COMPONENT BOARD CONSISTING OF:
45	I I, 13, 18, 20R	4	M6986 159F 10	RESISTOR 1000 OHMS, lb
46	10R	1	M6986 159E IO	RESISTOR 100 OHMS, 1W
47	12R	1	M6986 159D47	RESISTOR 47 OHMS, 1W
48	14R	1	M6986 159G 10	RESISTOR 10K OHMS, 1W
49	1 %	1	M6986 159F47	RESISTOR 4.7K OHMS, 1W
50	16R	1	M6986 159E22	RESISTOR 220 OHMS, 1W
51	21R	1	K8622212E33	RESISTOR 330 OHMS, 10W
52	10, 13C	2	44A337922J60	CAPACITOR 6MFD,200VDC
53	IO-21REC	12	448232019-004	RECTIFIER, IA, 400V
54	1Q	1	44B311057-0Q1	TRANSISTOR
55	2Q	1	44B312432-005	TRANSISTOR
56	8C	1	93385976 IO	CAPACITOR, .01MFD
57	FIRING BOARD	1*	3S7932MA199A6	COMPONENT BOARD CONSISTING OF:
58.	6R	1	M6986 159E 10	RESISTOR 100 OHMS, 1W
59	8D	1	44B232019-006	RECTIFIER, 1A, 600V
60	7D	1	44B2320 19-004	RECTIFIER, IA, 400V
61	1ZD	1	44B211517-101	ZENER DIODE 6.8V, 1W
62	4C	1	K9770378J IO	CAPACITOR , 1. OMFD
63		1	3S7932YA114A6	UNDER FREQUENCY LIMITER CONSISTING OF:

REVISIONS

711

PRINTS TO

* COMPONENT BOARDS SHOWN (*) MAY BE PURCHASED AS A COMPLETE UNIT OR COMPONENTS AS LISTED.

MADE BY Eleanor Ellison 8/10/76

ISSUED AUG 26 1976

APPROVALS

DRIVE SYSTEMS PROD. DEPT. DIV OR DEPT.

WAYNESBORO, VA. LOCATION

44A376059

CONT ON SHEET 3 SH NO. 2

CODE IDENT NO.

FF-803 WF (2-76) PRINTED IN U.S.A.

GENERAL ELECTRIC

44A376059

REV NO.

TITLE

CONT ON SHEET FNL. SH NO. 3

PRINCIPAL RENEWAL PARTS LIST FOR 3S7932SA112A1

CONT ON SHEET SH NO.

FIRST MADE FOR

HTEM NO.	CIRCUIT SYMBOL	QTY.	CATALOG NUMBER	DESCRIPTION OF PART
64	2T	1	44B314041-001	TRANSFORMER
65	3SX	1	44B214554-G01	REACTOR
66	5Q	1	2N3767	TRANSISTOR
67	14C	1	44B216128-010	CAPACITOR,20MFD,600VD
68	8TB	1	22006	TERMINAL BOARD
69		1*	3S7932MA215A5	UNDERFREQUENCY COMP. BOARD CONSISTING OF:
70	4ZD	1	44B211517-017	ZENER DIODE,24V,1.5W
71	22-29 REC	8	4482320 19-004	RECTIFIER IA, 400V
72	22R	1	K8622212F33	RESISTOR,3300 OHMS,10
73	23R	1	M6986 159E22	RESISTOR,220 OHMS,1W
74	24,25R	1	M6986 159E10	RESISTOR, 100 OHMS, 1W
75	26R	1	M6986159H10	RESISTOR,100K OHMS,1W
76	15C	1	K9774719K10	CAPACITOR 10MFD,50VDC
77	4Q	1	2N657A	TRANSISTOR

REVISIONS

1 /R. MORRIS

711

PRINTS TO

MADE BY
J. A. Ellison 8/10/76

ISSUED
AUG 26 1976

APPROVALS

DRIVE SYSTEMS PROD DEPT. DIV OR DEPT.
WAYNESBORO, VA.
LOCATION

44A376059

CONT ON SHEET FNL . SH NO. 3

CODE IDENT NO.

FF-803 WF (2-76)

GENERAL ELECTRIC

44A376060

CONT ON SHEET 2 SH NO. 1

REV NO.

TITLE

PRINCIPAL RENEWAL PARTS LIST FOR
3S7930SA113A1

CONT ON SHEET SH NO.

FIRS MADE FOR

ITEM NO.	CIRCUIT SYMBOL	QTY.	CATALOG NUMBER	DESCRIPTION OF PART	REVISIONS
1	2X	1	44B376804-G01	REACTOR	
2	SX	1	448374 1 IO-GO 1	REACTOR	
3	1PT,2PT	2	44B311312-G01	TRANSFORMER	
4	SCR	1	44B313954-002	RECTIFIER,180A,1300V	
5	1,2,3D	3	44A319818-006	RECTIFIER,250A,1200V	
6	4,5D	2	44A319818-106	RECTIFIER,250A,1200V	
7	IT	1	44B334557-G01	TRANSFORMER	
8	1,2,3C	3	28F5158FC	CAPACITOR,10MFD,2000V	
9	4C	1	28F5156FC	CAPACITOR,5MFD,2000V	
10	3R	1	44A333454-D15	RESISTOR,15 OHMS,225W	
11	IX	1	44B214515-G01	REACTOR	
12	1,2FU	2	933B578P17	FUSE 100A,600V	
13	3FU	1	K9774701P2	FUSE 3A,600V	
14	FOR 3FU	1	TC8050	FUSE BLOCK	
15	2R	1	44A333458-C20	RESISTOR,2.0 OHMS,225W	
16	6D	1	IN2 160	RECTIFIER 25A,600V	
17	FOR 6D,4R,5R	3	28 1A482P4	STANDOFF INSUL.	
18	1CR	1	CR205N 004	CONTACTOR	
19	THY	1	9RV6A2	THYRITE	
20	FOR POWER INPUT	8	44A31-1988-001	INSULATOR	
21	1,2TB	3	220 10	TERMINAL BOARD	
22	1SW	1	506 30	SWITCH	
23	4R	1	44A33345 I-G IO	RESISTOR, 10K,25W	
24	5R	1	K8622212E33	RESISTOR, 330 OHMS, 10W	
25	FOR 4R	2	44A339057-001	MICA WASHER	
26	FOR 2R,3R	4	44A339057-003	MICA WASHER	
27	AC REGULATOR	1	3S7932MA196C2	AC REGULATOR CONSISTING OF:	
28	2P, 5P	2	44A337916-104	POTENTIOMETER,500 OHMS 2W	
29	3P	1	M9729433F50	POTENTIOMETER, 5000 OHMS,2W	
30	1P	1	44A3379 16- 103	POTENTIOMETER 250 OHMS, 2W	
31	4P	1	M9729350025	POTENTIOMETER 25 OHMS, 25W	
32	3X	1	44B211589-G01	D. C. REACTOR	
33	11,12C	2	44A370874-K25	CAPACITOR,25MFD,200VDC	

711

PRINTS TO

MADE BY W. L. Shifflett 8-26-76
ISSUED AUG 27 1976

APPROVALS

DRIVE SYSTEMS PROD. DEPT.
WAYNESBORO, VA.

DIV OR DEP44A376060

LOCATION CONT ON SHEET 2 SH NO. 1

CODE IDENT NO.

FE-803 W F (2-76)

GENERAL ELECTRIC

44A376060

CONT ON SHEET 3 SH NO. 2

EV
EO.

TITLE
PRINCIPAL RENEWAL PARTS LIST FOR
3S7930SA1.13A1

CONT ON SHEET SH NO.

FIR MADE FOR

ITEM NO.	CIRCUIT SYMBOL	QTY.	CATALOG NUMBER	DESCRIPTION OF PART
34	3Q	1	2N3215	TRANSISTOR
35	2ZD	1	44A337971-017	ZENER DIODE,24V,10W,5%
36	3ZD	1	44A337972-011	ZENER DIODE,24V,50W,5%
37	FOR 2,3ZD H.S.	4	281A482P2	STANDOFF INSULATOR
38	10TB	1	220 10	TERMINAL BOARD
39	7TB	1	22008	TERMINAL BOARD
40	12TB	1	22003	TERMINAL BOARD
41	17R	1	K86222 13075	RESISTOR 75 OHMS,20W
42	17D	1	4482320 19-004	RECTIFIER IA, 400V
43	15R	1	44A33725 I-E IO	RESISTOR 100 OHMS,2W
44	REGULATOR BOARC	1*	3S7932MA 19882	COMPONENT BOARD CONSISTING OF:
45	11,13,18,20R	4	M6986 159F 10	RESISTOR 1000 OHMS, 1W
46	10R	1	M6986 159E 10	RESISTOR 100 OHMS, 1W
47	12R	1	M6986 159047	RESISTOR 47 OHMS, 1W
48	14R	1	M6986 159G10	RESISTOR 10K OHMS, 1W
49	19R	1	M6986 159F47	RESISTOR 4.7K OHMS, 1W
50	16R	1	M6986 159E22	RESISTOR 220 OHMS, 1W
51	21R	1	K8622212E33	RESISTOR 330 OHMS, 10W
52	IO, 13C	2	44A337922J60	CAPACITOR 6MFD,200VDC
53	IO-2 1REC	12	4462320 19-004	RECTIFIER, IA, 400V
54	1Q	1	44B311057-001	TRANSISTOR
55	2Q	1	448312432-005	TRANSISTOR
56	8C	1	933B597G IO	CAPACITOR, .O 1MFD
57	FIRING. BOARD	1*	3S7932MA199A6	COMPONENT BOARD CONSISTING OF:
58	6R	1	M6986 159E 10	RESISTOR 100 OHMS, 1W
59	8D	1	4482320 19-006	RECTIFIER, 1A,600V
60	7D	1	4482 320 19-004	RECTIFIER, 1A,400V
31	1ZD	1	448211517-101	ZENER DIODE 6.8V,1W
32	4C	1	K9770378J10	CAPACITOR, 1. OMFD
33		1	3S7932YA114A6	UNDER FREQUENCY LIMITER CONSISTING OF:

REVISIONS

711

PRINTS TO

* COMPONENT BOARDS SHOWN (*) MAY BE PURCHASED AS A COMPLETE UNIT OR COMPONENTS AS LISTED.

MADE BY H.L. Shiffler 8-26-76
ISSUED AUG 27 1976

APPROVALS J.P. Ruch

DRIVE SYSTEMS PROD. DEPT. DIV OR DEPT.
WAYNESBORO, VA. LOCATION

44A376060
CONT ON SHEET 3 SH NO. 2

FF-803 WF (2-76)
PRINTED IN U.S.A.

CODE IDENT NO.

GENERAL ELECTRIC

44A376060

REV NO.

CONT ON SHEET FNL. SH NO. 3

TITLE

PRINCIPAL RENEWAL PARTS LIST FOR
3S7932SA113A1

CONT ON SHEET SH NO. FIRST MADE FOR

ITEM NO.	CIRCUIT SYMBOL	QTY.	CATALOG NUMBER	DESCRIPTION OF PART
64	2T	1	44B314041-001	TRANSFORMER
65	3SX	1	44B214554-G01	REACTOR
66	5Q	1	2N3767	TRANSISTOR
67	14c	1	44B216128-010	CAPACITOR,20MFD,600VDC
68	8TB	1	22006	TERMINAL BOARD
69		1*	3S7932MA215A5	UNDERFREQUENCY COMP. BOARD CONSISTING OF:
70	4ZD	1	448211517-017	ZENER DIODE,24V,1.5W
71	22-29 REC	8	4482320 19-004	RECTIFIER 1A, 400V
72	22R	1	K86222 12F33	RESISTOR,3300 OHMS,10W
73	23R	1	M6986 159E22	RESISTOR,220 OHMS, 1W
74	24,25R	1	M6986 159E10	RESISTOR, 10Ω OHMS, 1W
75	26R	1	M6986 159H10	RESISTOR,100K OHMS, 1W
76	15C	1	K9774719K10	CAPACITOR 10MFD,50VDC
77	4Q	1	2N657A	TRANSISTOR

REVISIONS

R. MORRIS

711

PRINTS TO

MADE BY H.A. Elliott 8-26-76

ISSUED AUG 27 1976

APPROVALS

DRIVE SYSTEMS PROD. DEPT. DIV OR DEPT.
WAYNESBORO, VA. LOCATION

44A376060

CONT ON SHEET FNL. SH NO. 3

CODE IDENT NO.

FF-803 WF (2-76)
PRINTED IN U.S.A.

GENERAL ELECTRIC COMPANY ● DRIVE SYSTEMS DEPARTMENT
CONTROL DEVICES OPERATION ● WAYNESBORO, VIRGINIA 22980

GENERAL ELECTRIC

3/77 (30)